

Mini Project Report
ON
Travel Discovery & Social Sharing Platform



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CERTIFICATE

This is to certify that the project titled TRAVEL DISCOVERY & SOCIAL SHARING is the bona fide work carried out by Mr/MS. ITALIYA RIDHAM SURESHBHAIR (ENo. 25MCA30035) JAMBUCHA SNEHABEN MAGANBHAI (ENo. 25MCA30040) GAGLIYA YASH JAGADISHBHAI (ENo. 25MCA30030) ABHINAV KUMAR JHA (ENo. 25MCA30043) a student of LDRP Institute of Technology and Research MCA Department, during the academic year 2025 - 26, in partial fulfillment of the requirements for the award of the degree of MCA Semester 1 (Mini Project-I). 'कर भला होगा भला'

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Head of Department

Abstract

The Travel Discovery & Social Sharing Platform is an innovative web based application designed to simplify the process of exploring, planning, and sharing travel experiences. The platform aims to serve as a one stop destination for travelers to discover popular tourist attractions, hidden gems, and trending locations worldwide, eliminating the need to visit multiple travel websites.

The system allows users to create and share travel content, including photos, videos, blogs, and reviews, providing authentic insights into destinations from real travelers. Each user can maintain a personal travel diary, where their journeys, itineraries, and experiences are showcased, promoting an engaging and informative travel community. The interactive community feed encourages social connections among travelers, enabling features such as likes, comments, follows, and participation in group trips or travel challenges.

To enhance the discovery experience, the platform integrates smart recommendation algorithms that tailor travel suggestions based on user interests, budget, location, and past travel activities. Additionally, map integration enables real-time exploration by displaying nearby attractions along with travel routes, estimated costs, and travel times. As a potential enhancement, an AI powered travel planning assistant can further optimize itineraries by analyzing user preferences and time constraints.

This project not only addresses the challenges faced by modern travelers in finding reliable and personalized travel information but also promotes a collaborative digital community centered around authentic travel experiences. The Travel Discovery & Social Sharing Platform aims to redefine how people explore the world turning travel discovery into a shared, social, and smarter experience.

Acknowledgement

It gives me immense pleasure to present this project report on “**Travel Discovery & Social Sharing Platform**”, which has been a significant learning experience and an opportunity to apply theoretical knowledge to practical implementation.

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Problem Definition

In today's digital age, travelers rely heavily on multiple online sources such as travel blogs, video platforms, review sites, and social media to plan their trips and explore new destinations. However, the fragmentation of travel information across different platforms often leads to confusion, time consumption, and unreliable decision-making. Users struggle to find authentic, experience based insights about destinations, local attractions, and real travel experiences.

Existing travel platforms primarily focus on booking services or destination listings rather than creating a community driven discovery environment. There is a noticeable gap in platforms that allow travelers to both explore and socially engage, where they can not only find destinations but also share their own experiences, itineraries, and travel stories.

Moreover, travelers often lack personalized recommendations tailored to their interests, location, and budget. The absence of an integrated system that combines driven suggestions, social interaction, and map based exploration makes travel discovery inefficient and less engaging.

Therefore, there is a strong need for a unified digital solution that brings together travel discovery, user generated content, social engagement, and intelligent recommendations all within a single platform. The **Travel Discovery & Social Sharing Platform** aims to fill this gap by enabling users to explore destinations, share authentic experiences, connect with fellow travelers, and receive smart travel suggestions in a community oriented environment.

1. SRS

1.1 Current System

At present, travelers depend on a wide range of disconnected digital platforms for different stages of their travel journey from discovery and planning to sharing experiences. Information about tourist destinations, accommodations, and travel activities is scattered across various websites, mobile apps, and social media platforms.

Users commonly visit websites like TripAdvisor, MakeMyTrip, Google Travel, and YouTube to research destinations, watch travel vlogs, or read reviews. For inspiration and travel stories, they often rely on Instagram or travel blogs, while navigation and route details are accessed through Google Maps or similar services.

However, there is no single, unified platform that combines these functionalities i.e., destination discovery, trip planning, social sharing, and real user recommendations. Each existing system addresses only one aspect of travel and lacks integration with others.

This leads to several inefficiencies:

- Users spend excessive time switching between different apps and websites.
- Travel decisions rely on incomplete or promotional content rather than genuine experiences.
- There is minimal social engagement or collaboration among travelers.

Thus, the current system is fragmented, time-consuming, and lacks a sense of community-driven discovery, making it difficult for travelers to find reliable, personalized, and experience-based travel information in one place.

1.2 Its Drawbacks

The existing travel information systems, though diverse and feature-rich in isolation, suffer from several significant drawbacks when viewed from the perspective of a modern traveler. These drawbacks affect the efficiency, reliability, and user experience during the process of discovering, planning, and sharing travel activities.

The major drawbacks of the current system are as follows:

1. Fragmented Information:

Travel-related data such as reviews, videos, maps, and recommendations are spread across multiple platforms, forcing users to switch between different sources to gather complete information.

2. Lack of Authenticity:

Many reviews and travel blogs are promotional or biased, making it difficult for users to find genuine, experience-based opinions from real travelers.

3. No Unified Travel Experience:

Existing applications either focus on booking services or destination discovery, but none combine travel exploration, content sharing, and social interaction in one system.

4. Limited Personalization:

Current platforms do not provide AI-driven or interest-based recommendations that adapt to the user's travel history, preferences, or budget.

5. Absence of Social Community:

There is no dedicated community feature that enables travelers to interact, collaborate, share experiences, or plan group trips collectively.

1.3 Proposed System and its features

The proposed system, **Travel Discovery & Social Sharing Platform**, is a unified digital application designed to bring together **travel exploration, user-generated content, social interaction, and smart recommendations** within a single, integrated environment. The primary goal of this system is to **simplify travel discovery** and enhance user engagement by combining authentic travel information with interactive community features.

Unlike existing platforms, this system enables users to **discover destinations, share personal experiences, interact with other travelers, and receive AI-powered suggestions** — all in one place. It eliminates the need for users to visit multiple websites or apps to plan a single trip.

The platform promotes a **community driven travel ecosystem**, where travelers can not only consume content but also contribute their own stories, tips, and itineraries. Through smart analytics and personalized algorithms, the system curates content and recommendations that match each user's preferences, interests, and travel history.

Key Features of the Proposed System :

- 1. Travel Discovery Hub:**

Explore trending destinations, local attractions, and hidden gems through curated content and real traveler contributions.

- 2. User Generated Content:**

Users can upload photos, videos, blogs, and reviews to share real experiences, helping others make informed travel decisions.

- 3. Personal Travel Diary:**

Maintain a digital log of visited places, trips, and itineraries that can be viewed and shared with others.

- 4. Interactive Community Feed:**

Like, comment, follow other travelers or group trips.

5. **Smart Recommendation Engine:**

AI-powered algorithms suggest destinations and activities based on user interests, location, budget, and travel patterns.

6. **Map Integration:**

View nearby attractions, routes, travel times, and estimated costs on an interactive live map for seamless navigation.

7. **Social Sharing & Engagement:**

Post travel updates, connect with fellow explorers, and build a network of like-minded travelers globally.

8. **AI Powered Planning Assistant (Optional Enhancement):**

Automatically generate optimized travel itineraries that balance user preferences, available time, and budget.

9. **Cross Platform Accessibility (Optional Enhancement):**

Accessible on both **web and mobile devices**, ensuring a smooth and consistent user experience across platforms.

10. **Secure User Authentication:**

Implements robust login, registration, and profile management features with role-based access control to ensure user data privacy.

1.4 Scope

The Travel Discovery & Social Sharing Platform aims to provide travelers with a comprehensive, interactive, and intelligent system that simplifies the process of discovering destinations, planning trips, and sharing travel experiences within a single platform. The system's scope extends across web and mobile environments, targeting both casual travelers and travel enthusiasts who wish to connect, explore, and inspire others.

This project focuses on integrating social networking features, content-sharing tools, and AI-based recommendation systems to create a seamless travel experience. Users can discover popular attractions, view authentic travel stories, and make informed decisions through real user-generated content. Additionally, the platform promotes a collaborative travel community, allowing travelers to interact, follow others, and participate in travel-related challenges or events.

From a technical perspective, the system will include secure authentication, cloud-based data management, location-based services, and real-time map integration for route and attraction visualization. It will also provide personalized recommendations by analyzing user preferences, interests, and travel history using intelligent algorithms.

The platform can later be expanded with advanced features such as:

- AI-powered itinerary planning and budgeting tools.
- Integration with travel booking APIs (hotels, flights, and transport).
- Augmented Reality (AR) exploration of destinations.

In summary, the scope of this project covers the design, development, and implementation of a web-based and mobile-compatible travel application that enables:

- Destination Discovery through user-generated content.
- Social Connectivity among travelers worldwide.
- Smart Recommendations powered by AI.

The system will serve as a one-stop travel platform, providing convenience, reliability, and engagement to users, while establishing a strong foundation for future technology.

1.5 Hardware and Software Specification

Table 1.5.1 Hardware Requirements (For Web Based System)

Sr. no.	Hardware Component	Minimum Requirement	Recommended Requirement
1.	Client Device (User)	Any device with browser	Laptop / Smartphone
2.	Processor (Dev/Server)	Dual Core CPU	intel i5 / Ryzen 5
3.	RAM	4 GB	8–16 GB
4.	Internet Connection	2 Mbps	10+ Mbps stable

Table 1.5.2 Software Requirements (Web Application)

Sr. no.	Hardware Component	Version / Type
1.	Operating System	Windows / Linux / macOS
2.	Node.js Runtime	v18+
3.	Next.js Framework	v13+ (App Router)
4.	React	v18+
5.	NPM / Yarn	Latest
6.	Database	MongoDB 6.0+
7.	IDE	VS Code
8.	Browser	Chrome / Firefox / Edge
9.	Postman / Thunder Client	Latest
10.	Authentication Tools	bcrypt, JWT

1.6 Feasibility Study

A **Feasibility Study** is performed to evaluate the practicality and viability of developing the **Travel Discovery & Social Sharing Platform**. It ensures that the project can be successfully implemented with available resources, within the given timeframe, and that it meets user expectations effectively.

The feasibility analysis covers four main aspects **Technical, Operational, Economic, and Time** feasibility

1.6.1. Technical Feasibility

The proposed system is **technically feasible** as it utilizes modern, scalable, and open source technologies.

The application is designed to be developed using **React.js** for the frontend, **Node.js with Express.js** for the backend, and **MongoDB** as the database. These technologies are widely adopted, well-documented, and suitable for building a dynamic, real-time, and responsive platform.

Integration with **Google Maps API** will enable location-based services, while optional AI modules (using Python or TensorFlow.js) will provide smart recommendations for travelers. The hardware requirements are minimal — a system with basic specifications can run the development environment smoothly.

Hence, from a technical perspective, all tools and technologies required for system implementation are available, reliable, and feasible to use.

1.6.2. Operational Feasibility

The **Travel Discovery & Social Sharing Platform** is **operationally feasible** as it directly addresses the challenges faced by modern travelers in discovering, planning, and sharing travel experiences.

The system provides an intuitive and user-friendly interface where users can:

- Register and create personal profiles.
- Explore destinations and upload travel stories.
- Interact socially through likes, comments, and follows.
- Get travel recommendations based on interests and preferences.

Since the platform functions similar to familiar social media applications, users will find it easy to operate with minimal learning. Moreover, real-time interaction and social engagement will encourage active user participation, ensuring successful adoption.

1.6.3. Economic Feasibility

The system is **economically feasible**, as it is built using open-source frameworks and tools, significantly reducing the overall cost of development and deployment.

Cost considerations include:

- Hardware: Standard personal computers and mobile devices.
- Software: Open-source technologies (React.js, Node.js, MongoDB).
- Hosting: Cloud-based (AWS/Firebase) with minimal cost options.

Additionally, the project offers opportunities for **future monetization** through:

- Advertisements and sponsored posts.
- Premium user memberships for advanced features.
- Affiliate programs with travel agencies or hotel partners.

Thus, the project can be developed and maintained cost-effectively while offering strong potential for future revenue.

1.6.4. Time Feasibility

The estimated development period for this project is around **4–5 months**, which aligns well with the academic semester timeline.

Using an **Agile development model**, the work can be divided into sprints each covering requirement gathering, design, development, testing, and refinement phases.

1.6.5. Conclusion

Based on the above evaluation, the **Travel Discovery & Social Sharing Platform** is found to be **technically sound, operationally practical, economically viable, and time-efficient**.

Hence, the project is **feasible for successful development and deployment** with the available resources and within the academic duration.

1.7 Fact Finding Methods

Fact-finding is the process of gathering information, understanding user requirements, and analyzing the existing challenges to design an effective and user-friendly system. For the development of the **Travel Discovery & Social Sharing Platform**, several fact-finding techniques were used to collect relevant data, evaluate user expectations, and understand current travel-related digital behavior.

The following methods were adopted:

1.7.1. Observation

Direct observation was conducted to analyze how travelers currently use different platforms such as **TripAdvisor, Instagram, Google Maps, and YouTube** for exploring destinations and sharing experiences.

Through observation, it was found that users frequently switch between multiple apps, which creates inconvenience and leads to incomplete information gathering. This method helped identify gaps in existing systems and the need for an integrated solution.

1.7.2. Questionnaire Survey

A structured questionnaire was distributed among frequent travelers, students, and travel enthusiasts to understand their preferences, pain points, and expectations from a travel platform.

Key questions focused on:

- How users currently plan their trips.
- Which features they value most (e.g., map integration, reviews, travel blogs, recommendations).
- What problems they face while using current travel applications.

Survey results confirmed that users prefer a **community-driven platform** that combines discovery, planning, and social engagement in one place.

1.7.3. Interviews

Personal interviews were conducted with travel bloggers, tourism students, and software users to gain deeper insights into how digital tools influence travel decisions.

These interviews provided qualitative data regarding usability issues, missing features in existing platforms, and suggestions for improvement.

For example, users expressed interest in having **AI-based personalized suggestions** and the ability to **share travel itineraries** easily.

1.7.4. Research and Literature Study

Secondary research was carried out by reviewing travel-related articles, industry reports, and existing online platforms.

This helped in understanding global trends in **travel technology, social media integration, and AI-based recommendations**, which guided the system's conceptual and technical design.

1.7.5. Brainstorming Sessions

Team brainstorming sessions were conducted among the project group members to discuss gathered data, share creative ideas, and finalize key modules of the system such as:

- User authentication and profile management.
- Travel feed and content sharing.
- Recommendation engine design.
- Map and navigation integration.

1.8 Process Model

A **Process Model** defines the approach, methodology, and structured framework used during the development of a software system. It ensures systematic progress from requirement gathering to final deployment, with well-defined stages for design, coding, testing, and maintenance.

For the **Travel Discovery & Social Sharing Platform**, the **Agile Software Development Model** is selected, as it supports **iterative development**, **continuous feedback**, and **incremental delivery** all crucial for building a dynamic and user-centric platform.

1.8.1 Agile Model Overview

The **Agile Model** emphasizes flexibility, adaptability, and close collaboration between developers and stakeholders. Instead of building the entire system in one go, the project is divided into multiple short cycles known as “**Sprints**”, each delivering a functional module of the application.

This approach ensures that changes in user requirements can be easily accommodated during development, making it ideal for a project that combines travel content, social features, and AI-driven recommendations.

Furthermore, Agile promotes a **user-centered design philosophy**. The focus is not just on completing technical tasks but on creating real value for users. By releasing functional prototypes at the end of each sprint, the development team can collect real user feedback and use it to guide design enhancements.

This ensures that the final platform not only meets functional requirements but also delivers an engaging and intuitive user experience.

Key Characteristics of Agile Model:

1. Incremental Development:

The project is divided into smaller deliverables (modules), each completed and tested in short time frames.

2. Continuous Feedback:

User and stakeholder feedback is gathered after each sprint to refine features and improve usability.

3. Collaborative Approach:

All team members — developers, testers, designers — work collaboratively, ensuring integration and consistency.

4. Adaptability to Change:

New requirements or design modifications can be implemented in future iterations without disrupting progress.

5. Testing in Every Phase:

Quality assurance and testing occur continuously throughout development, not just at the end.

Agile Model Phases:

1. Requirement Gathering & Analysis:

Identify functional and non-functional requirements through user research, questionnaires, and brainstorming.

2. Planning:

Define sprint goals, assign responsibilities to team members, and set deliverables for each iteration.

3. Design:

Prepare system architecture, database schema, UI wireframes, and API structure for the sprint module.

4. Development:

Code the specific features planned for that sprint (e.g., login module, feed system, map integration).

5. Testing:

Perform unit, integration, and user testing to ensure the module functions correctly and meets quality standards.

6. Review & Feedback:

Demonstrate the working feature to users/stakeholders, gather feedback, and make improvements if required.

7. Deployment & Maintenance:

After all sprints are completed, deploy the system and provide regular updates and bug fixes.

Advantages of Using Agile for This Project:

- Faster delivery of working modules.
- High flexibility for feature enhancement (e.g., adding AI or map integrations later).
- Continuous improvement through real-time feedback.
- Improved collaboration among team members.
- Ensures a user-centered and high-quality final product.

Conclusion

The **Agile Process Model** provides the best framework for developing the **Travel Discovery & Social Sharing Platform**. Its iterative and user-focused nature ensures that the project evolves with user expectations and delivers a reliable, engaging, and scalable product within the defined timeline.

1.9 Development strategy and Type of System

The development of the **Travel Discovery & Social Sharing Platform** follows a **modular and iterative strategy**, ensuring that each component of the system is designed, developed, tested, and integrated efficiently. The development strategy is based on the **Agile methodology**, allowing the team to deliver functional modules incrementally while incorporating user feedback throughout the process.

This strategy ensures that system development remains **flexible, adaptive, and aligned** with the evolving needs of users and the project objectives. The project is divided into three major development layers — **Frontend (Client-Side)**, **Backend (Server-Side)**, and **Database (Data Layer)** — each developed in parallel with proper synchronization and version control using **Git and GitHub**.

1.9.1 Development Strategy

a) Frontend Development (Client-Side)

The frontend of the system focuses on providing an **intuitive, interactive, and visually appealing interface** for users. It is developed using **React.js**, a popular JavaScript library that enables the creation of dynamic and responsive single-page applications (SPAs).

Key aspects of the frontend include:

- A **clean and user friendly interface** that allows easy navigation through travel posts, destinations, and maps.
- **Responsive design** that ensures compatibility devices (desktop).
- **Integration of APIs** for displaying maps, routes, and nearby attractions in real time.
- **Reusable UI components** for consistent design and faster development.

Technologies used :

NEXT JS, SCSS, JavaScript (ES6).

b) Backend Development (server side)

The backend handles the **business logic, authentication, and communication between the frontend and database**. It is developed using **Node.js** and **Express.js**, providing a lightweight and scalable environment suitable for modern web applications.

Key functionalities implemented in the backend include:

- **User Authentication & Authorization** (Sign up, Login, Profile Management).
- **Data Management** for posts, comments, likes, and travel diaries.
- **RESTful API Development** for smooth data exchange between client and server.
- **Integration of third party APIs** such as Google location.
- **Recommendation Engine** (optional enhancement) for personalized travel suggestions based on user preferences.

Technologies used:

NEXT JS, Node.js, Express.js, JWT Authentication, RESTful APIs.

c) Database Design and Management

The system uses a **NoSQL database (MongoDB)** due to its scalability, flexibility, and ability to handle large volumes of user-generated data efficiently. Each collection stores structured and semi-structured data such as user profiles, travel posts, likes, and comments.

Database modules include:

- **User Collection** – stores personal details, authentication data, and travel history.
- **Post Collection** – holds travel blogs, videos, reviews, and geolocation data.
- **Interactions Collection** – manages likes, comments, and follows for social engagement.

Technologies used:

MongoDB, Mongoose ORM.

d) Testing and Quality Assurance

Testing is conducted throughout the development cycle, ensuring that every module performs as expected. Both **White Box** and **Black Box Testing** are used to identify errors, performance bottlenecks, and security vulnerabilities. Tools such as **Postman**, **Jest**, and **Mocha** are used for testing APIs and modules.

e) Deployment and Maintenance

The system is designed to be **cloud-deployable**, allowing for easy scalability and global accessibility. Deployment can be done using cloud platforms like **AWS**, **Firebase**, or **Render**. The system architecture supports future maintenance, version updates, and the integration of advanced features like AI or AR.

1.9.2. Type of System

The **Travel Discovery & Social Sharing Platform** can be classified as:

1. Web Based System:

Accessible through browsers on any device, providing global availability and convenience to users.

2. Social Networking System:

Encourages interaction among users through likes, comments, follows, and sharing features fostering a travel community.

3. Content Management System (CMS):

Allows users to create, upload, and manage their travel content in the form of blogs, photos, and videos.

4. Recommendation System:

Suggests destinations, itineraries, and activities based on user preferences, location, and budget.

5. Location based System:

Uses map integration to provide real-time navigation, nearby attractions, and route planning.

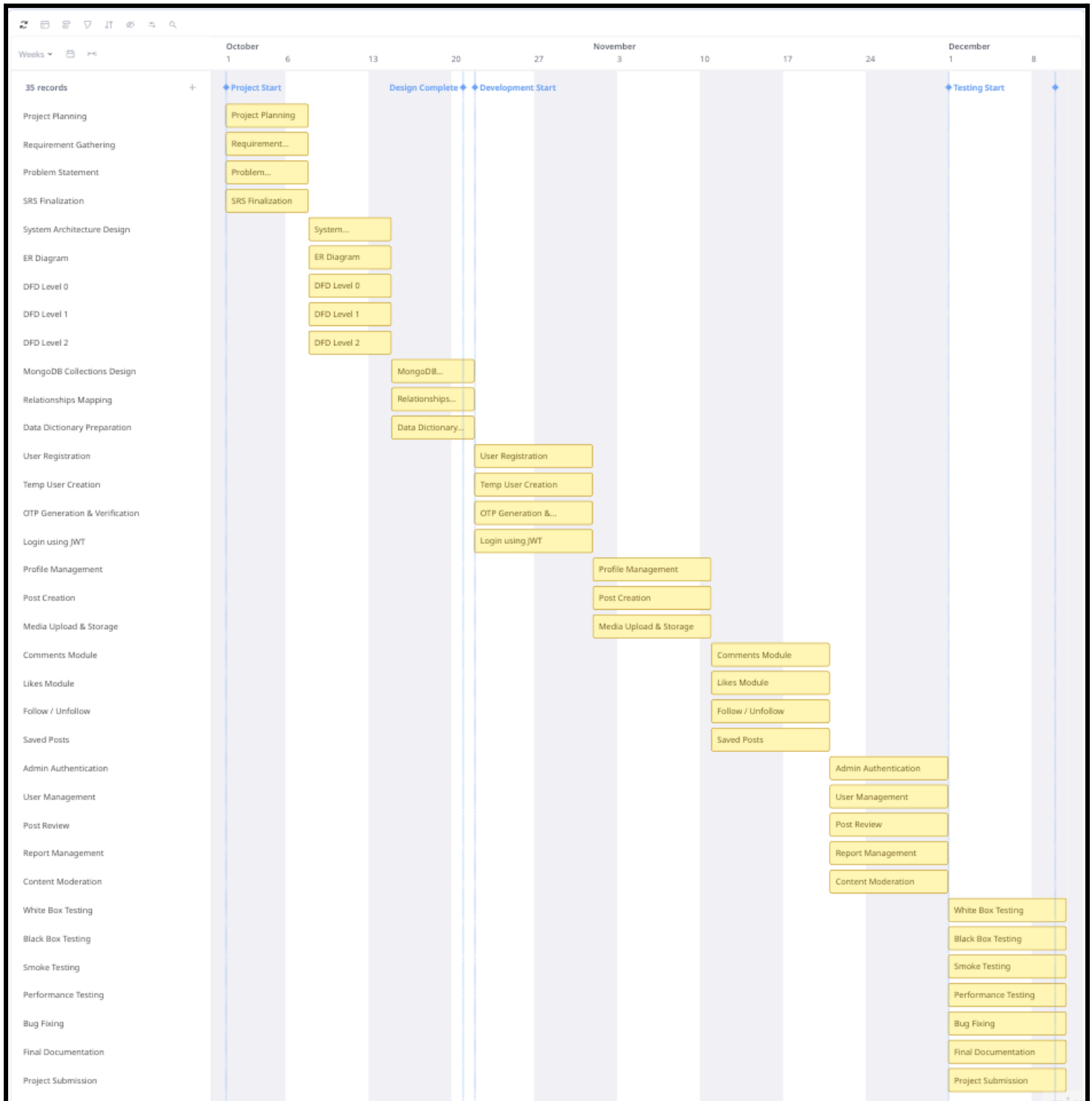
Conclusion

The adopted **development strategy** ensures a **structured, modular, and efficient** approach to building the system, while the chosen **type of system** reflects its multi-functional nature — combining travel exploration, social networking, and intelligent recommendations within a unified platform.

The **Travel Discovery & Social Sharing Platform** thus serves as a scalable, user-centric, and technologically advanced solution designed to redefine digital travel experiences.

2. Analysis

2.1 Gantt Chart (Estimated)



2.2 System Flow Chart

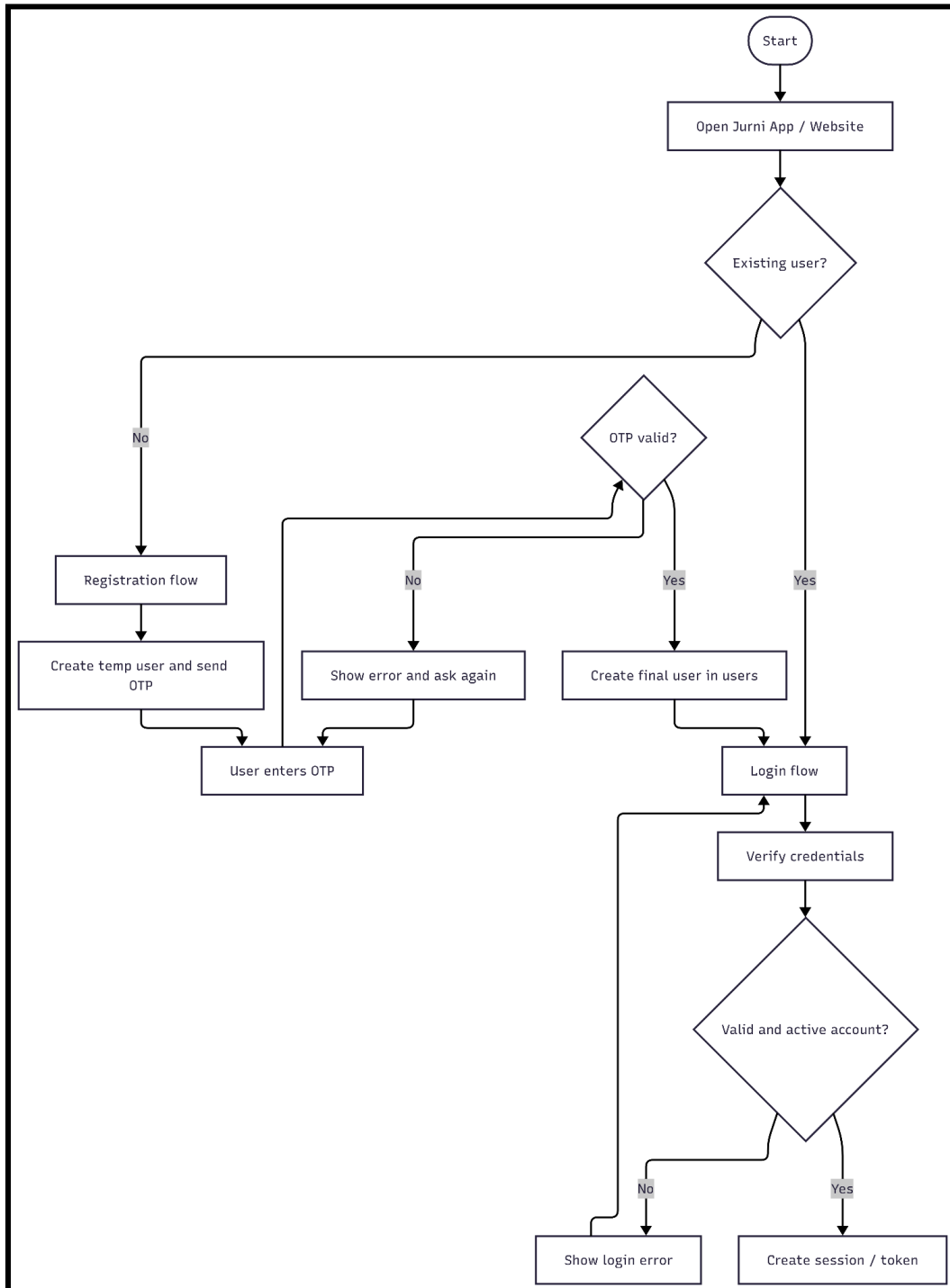


Fig. 2.2.1 System Auth Flow

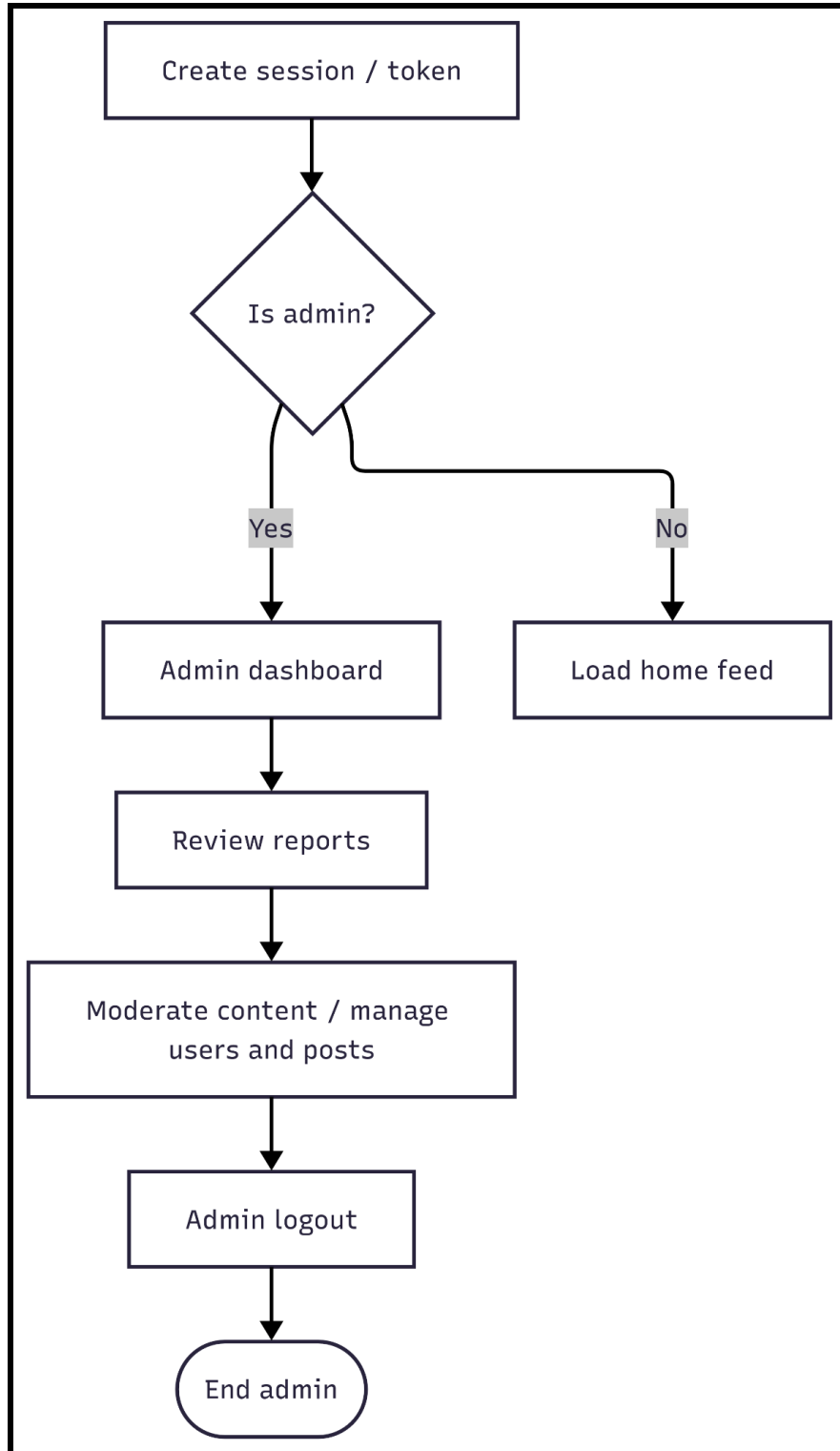


Fig. 2.2.2 System Admin/User Flow

Fig. 2.2.3 System Content Flow

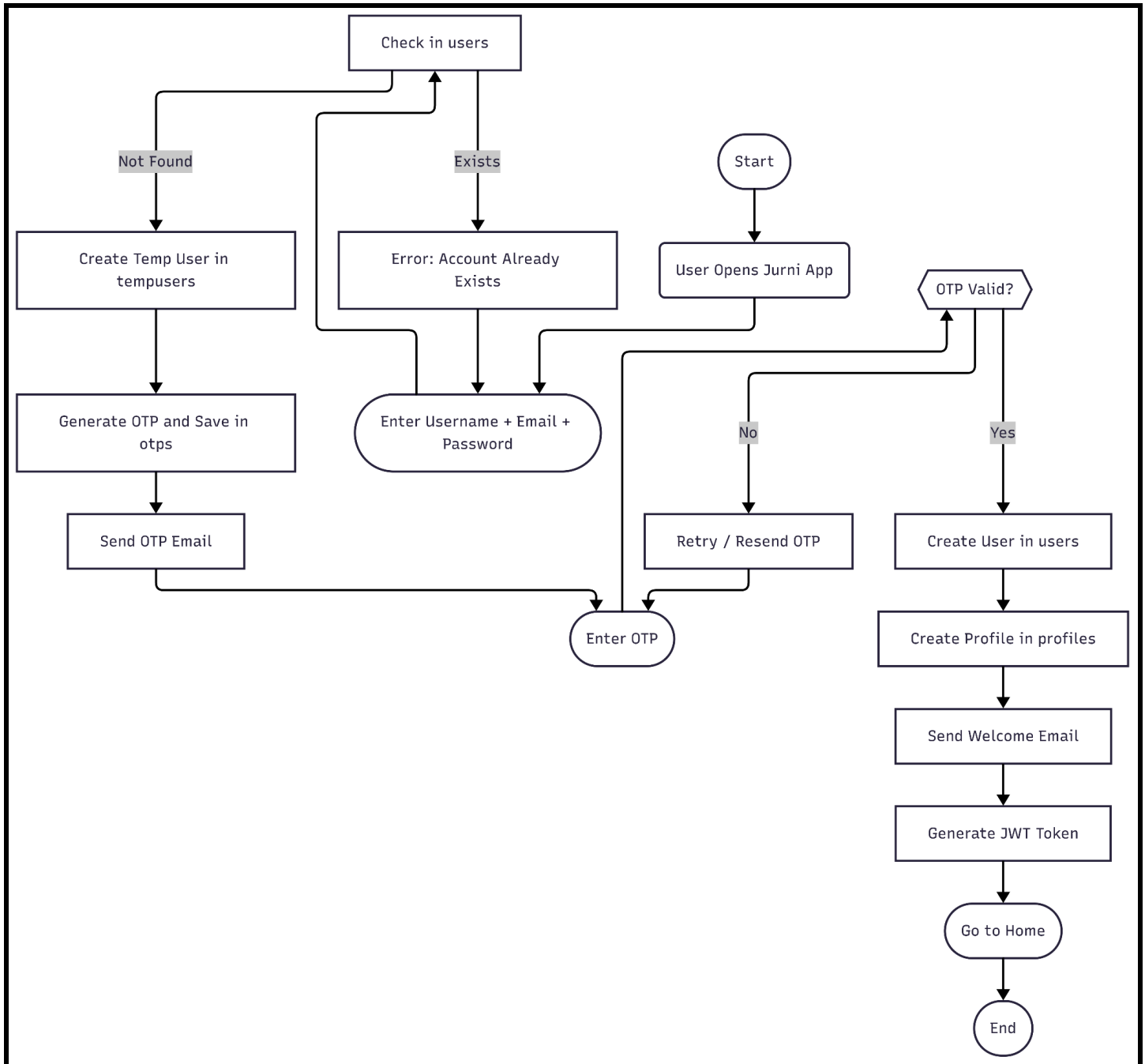


Fig. 2.2.4 System Registration Flow

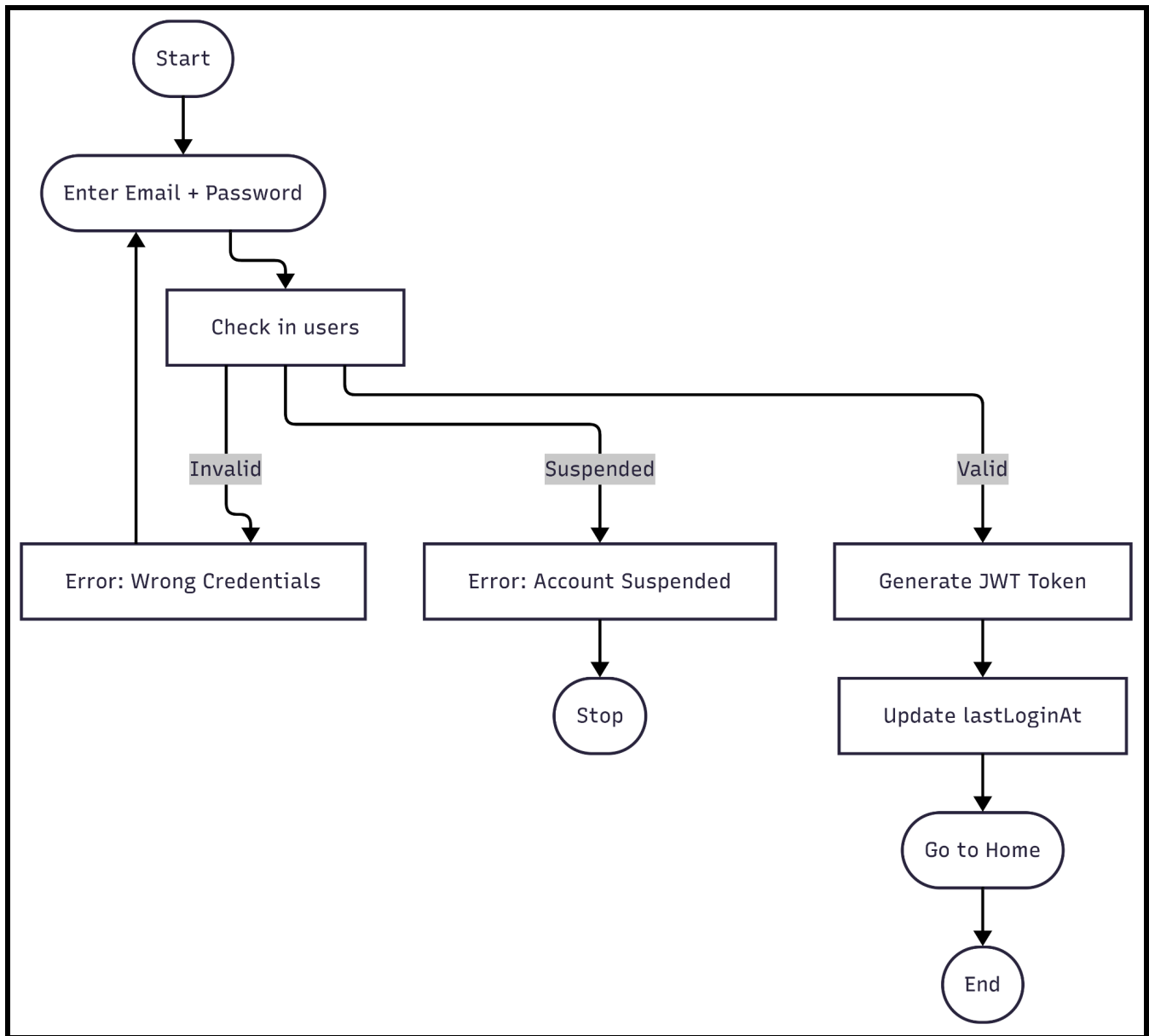


Fig. 2.2.5 System Login Flow

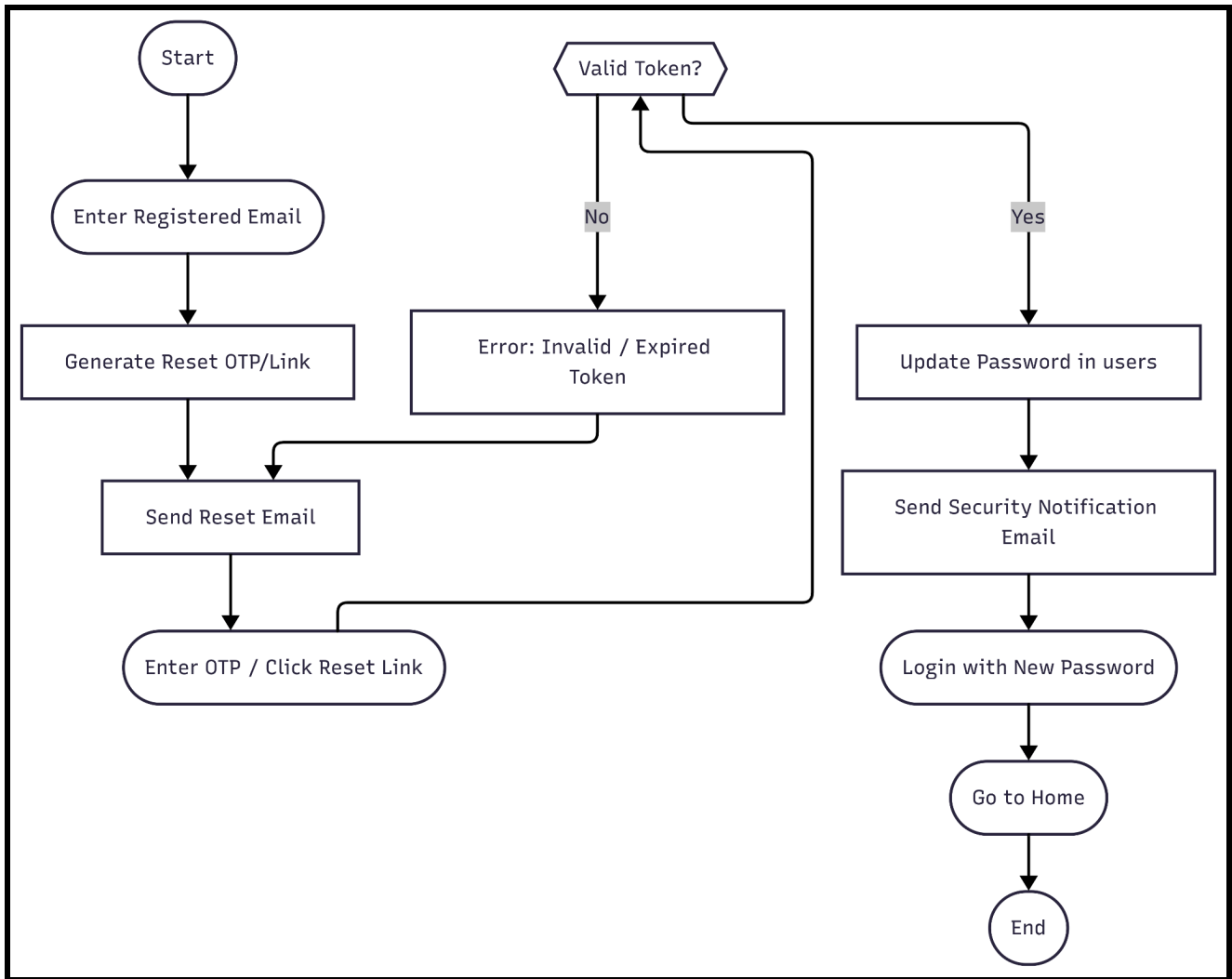


Fig. 2.2.6 System Reset Password Flow

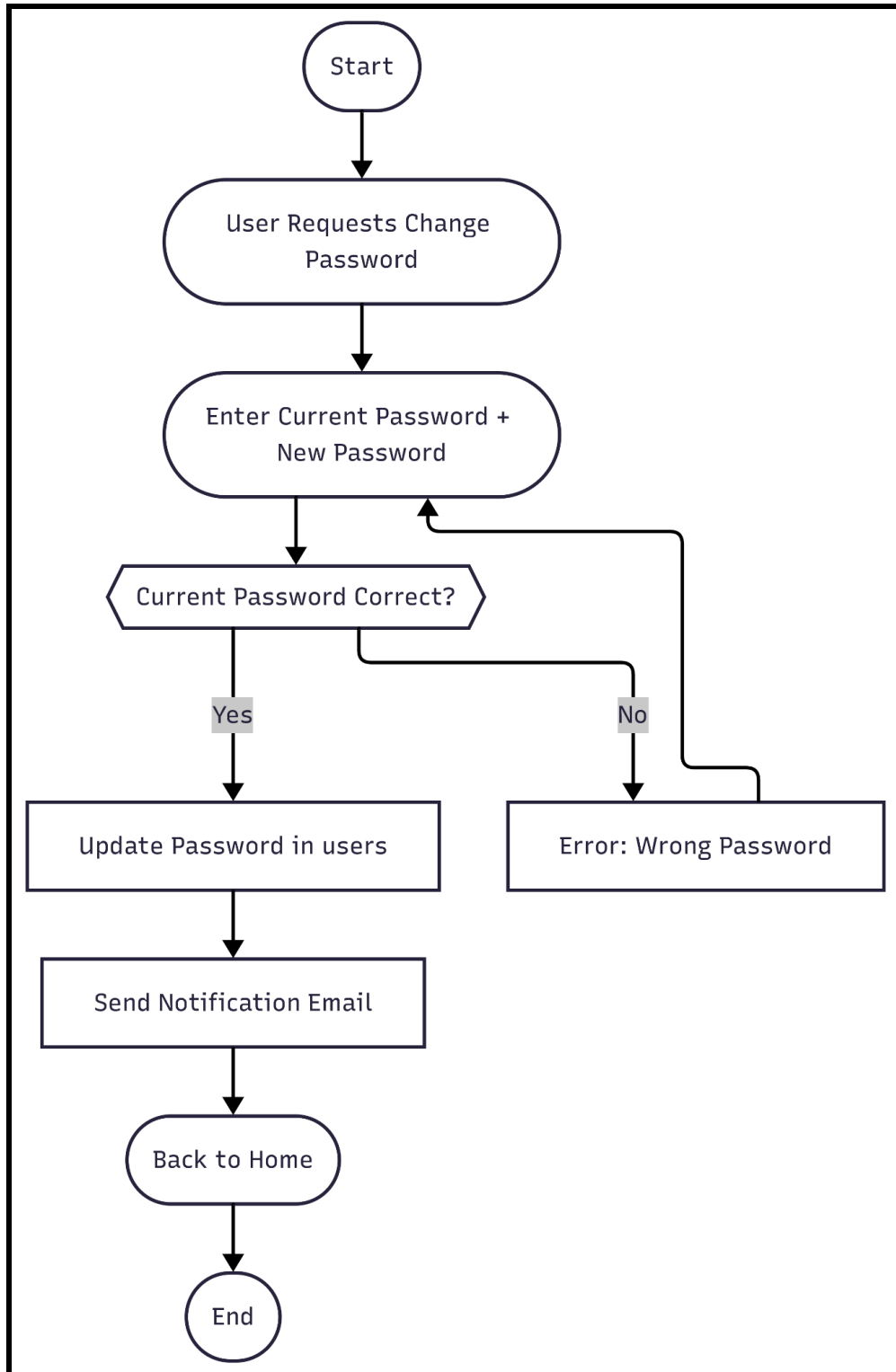


Fig. 2.2.7 System Change Password Flow

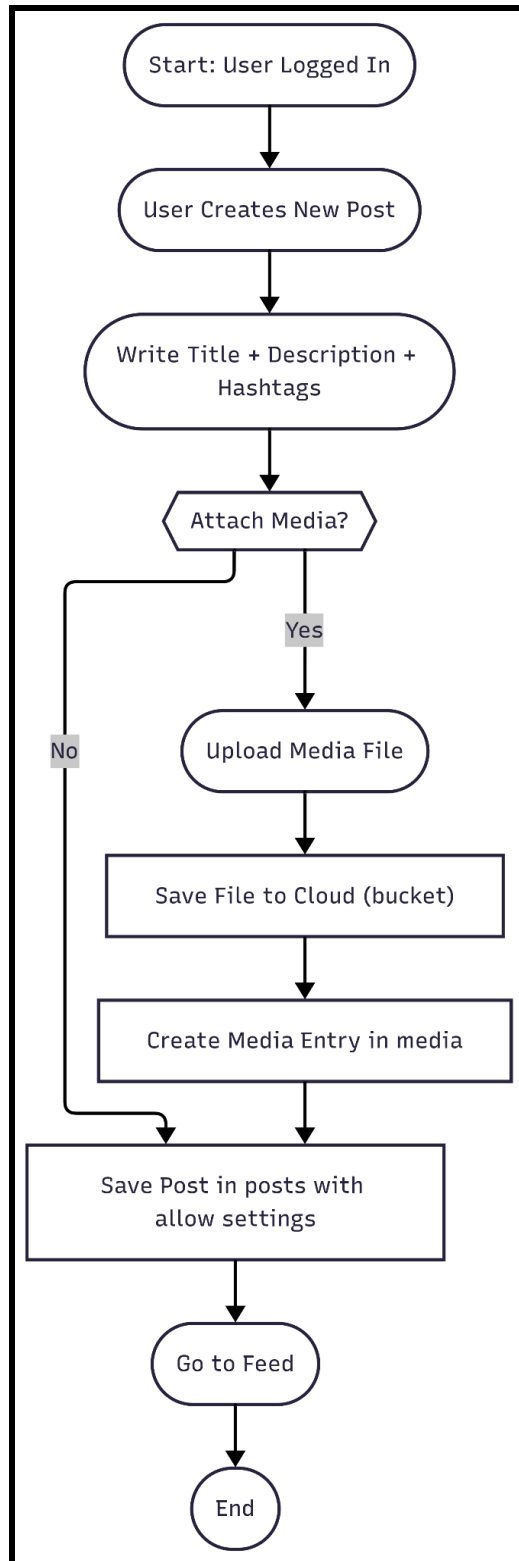


Fig. 2.2.8 System Create Content Flow

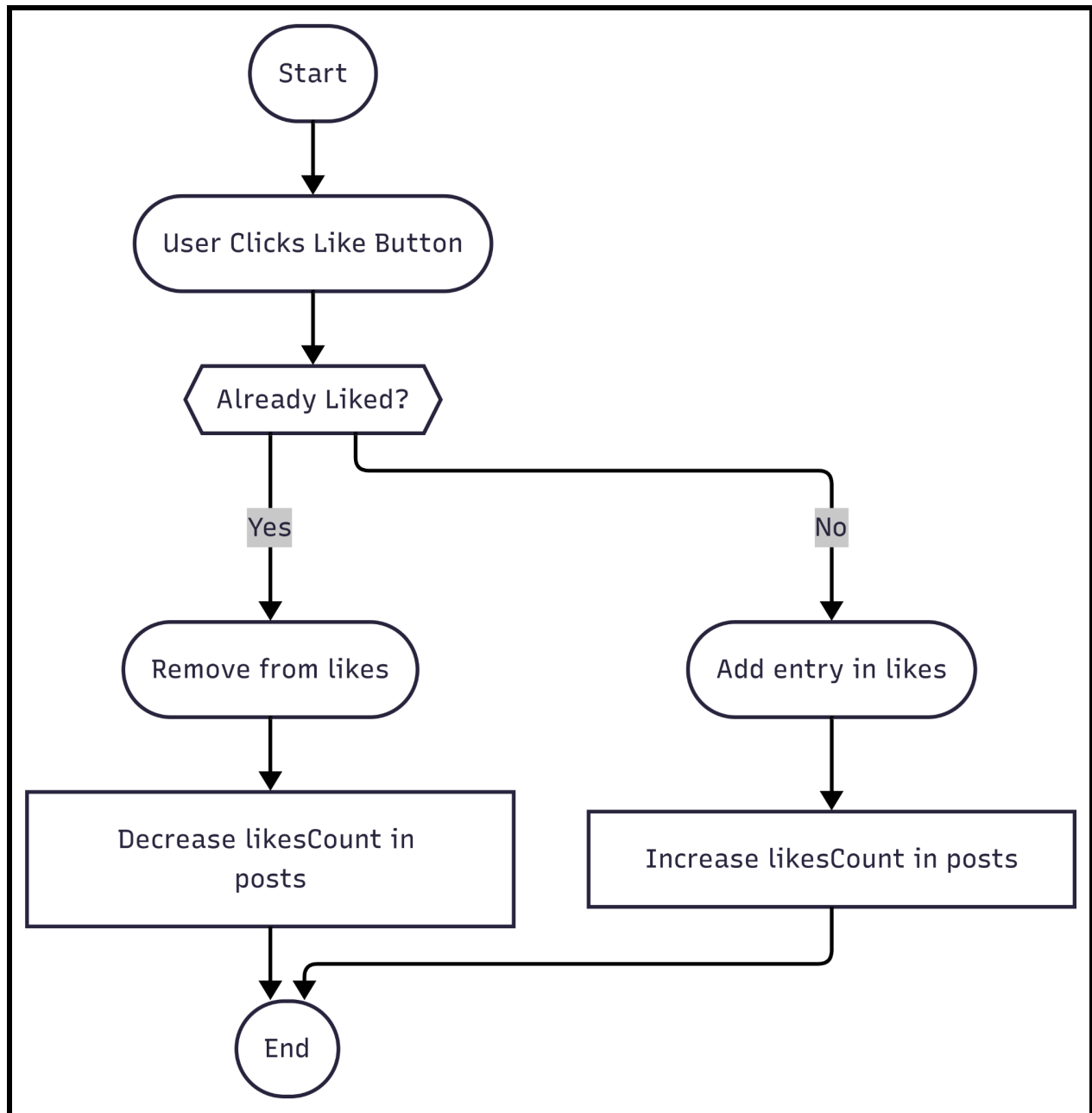


Fig. 2.2.9 System Like Unlike Flow

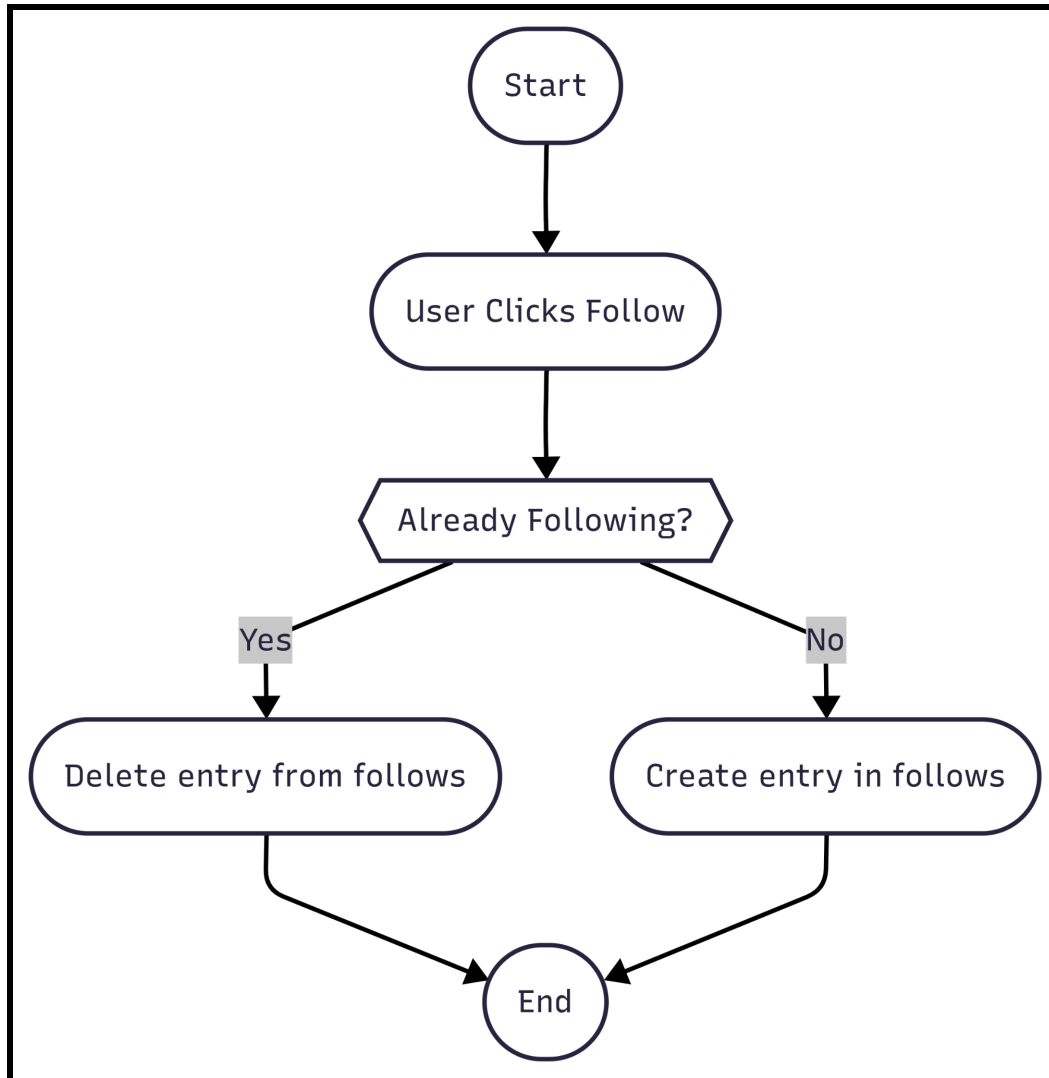


Fig. 2.2.10 System Follow Unfollow Flow

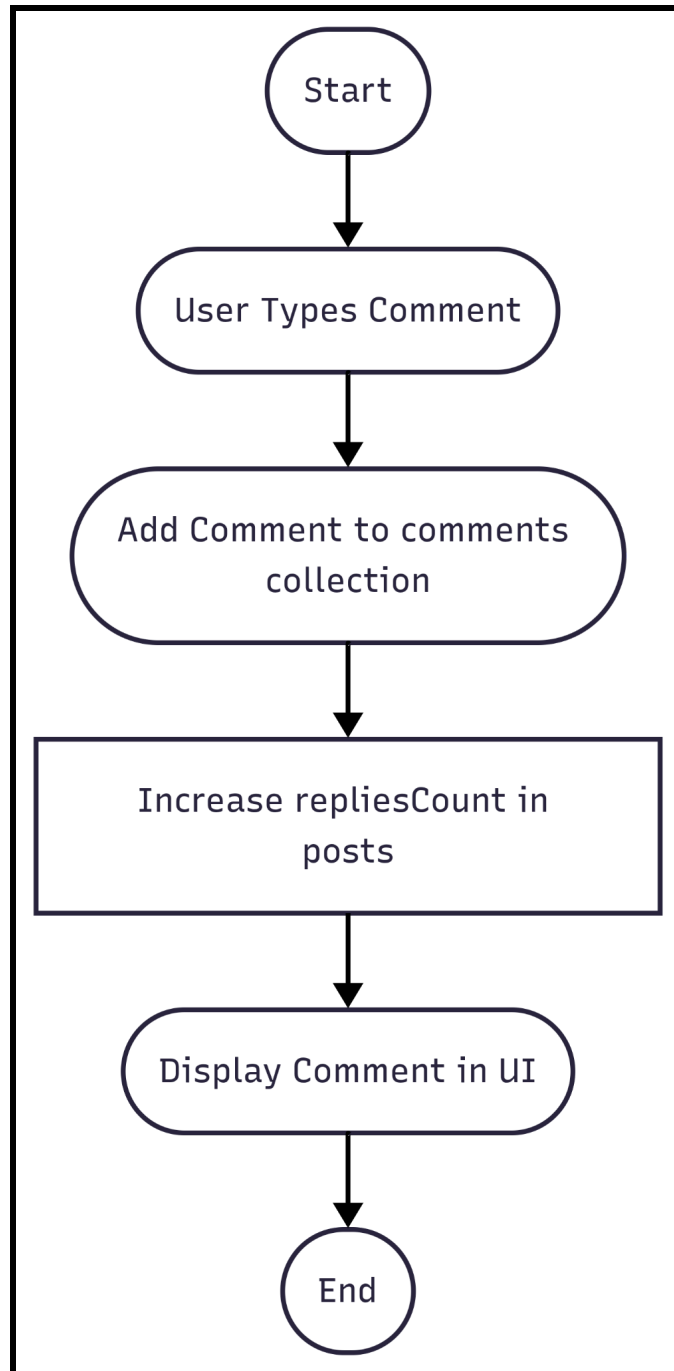


Fig. 2.2.11 System Comment Flow

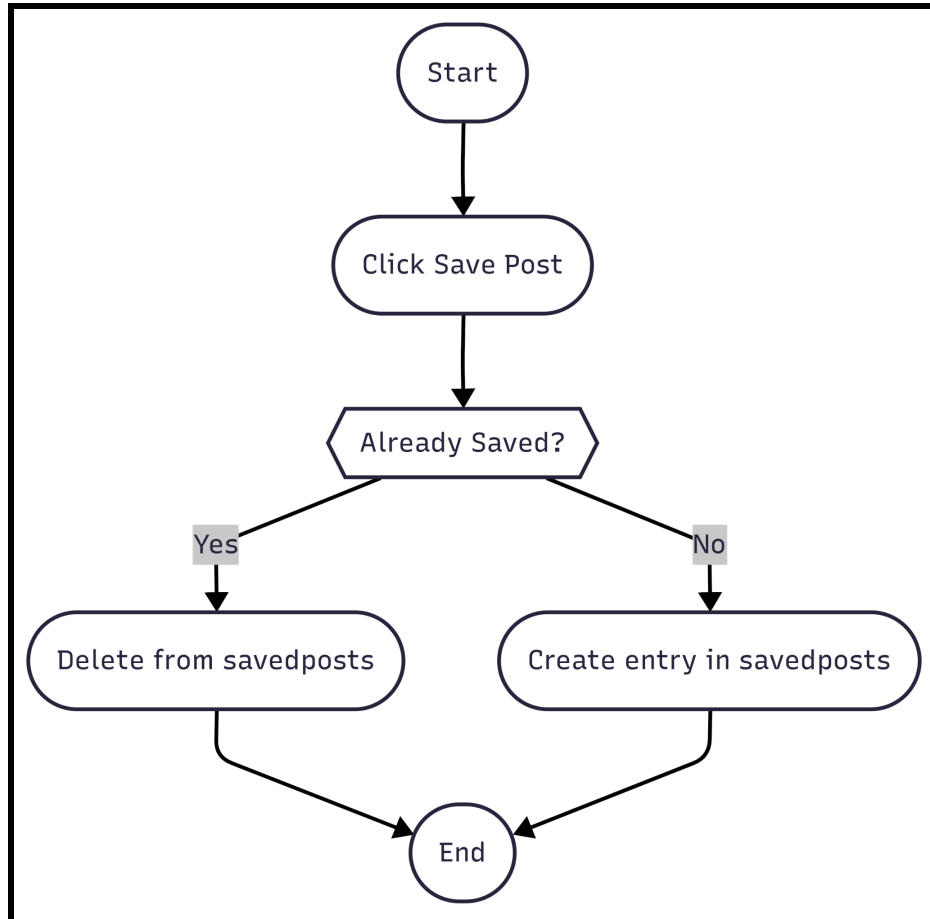


Fig. 2.2.12 System Save Flow

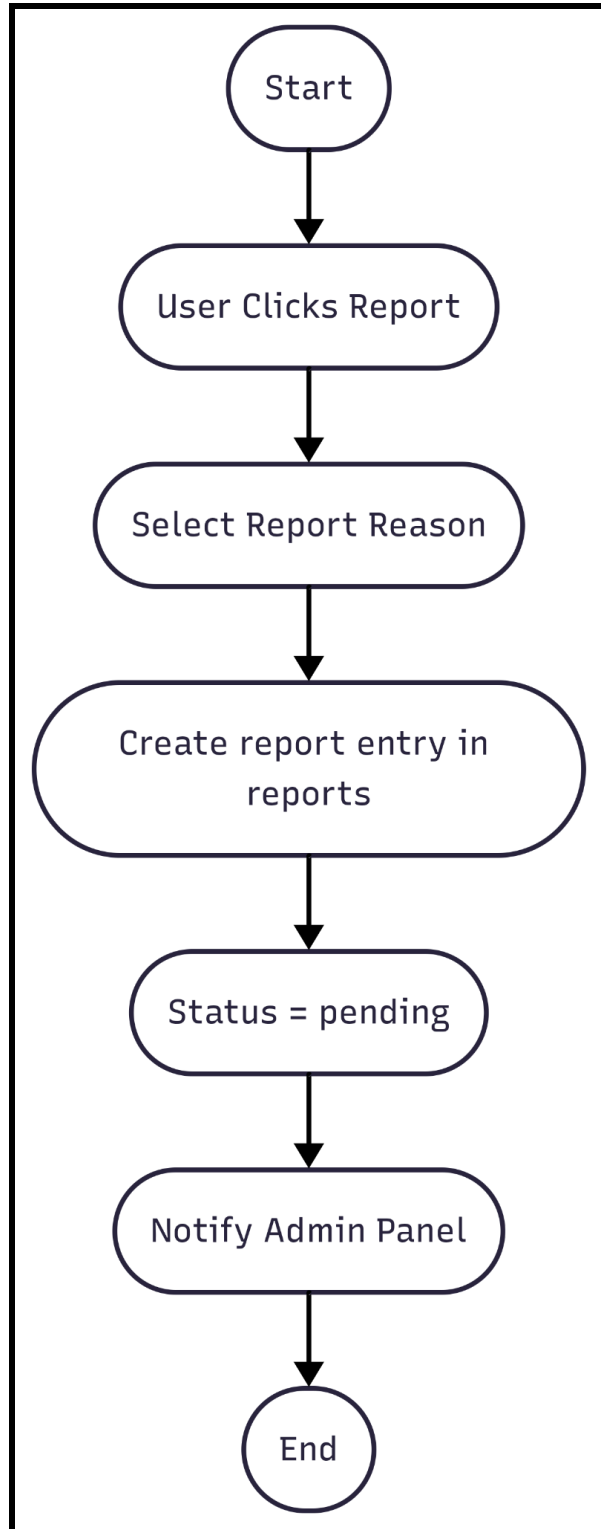


Fig. 2.2.13 System Report Flow

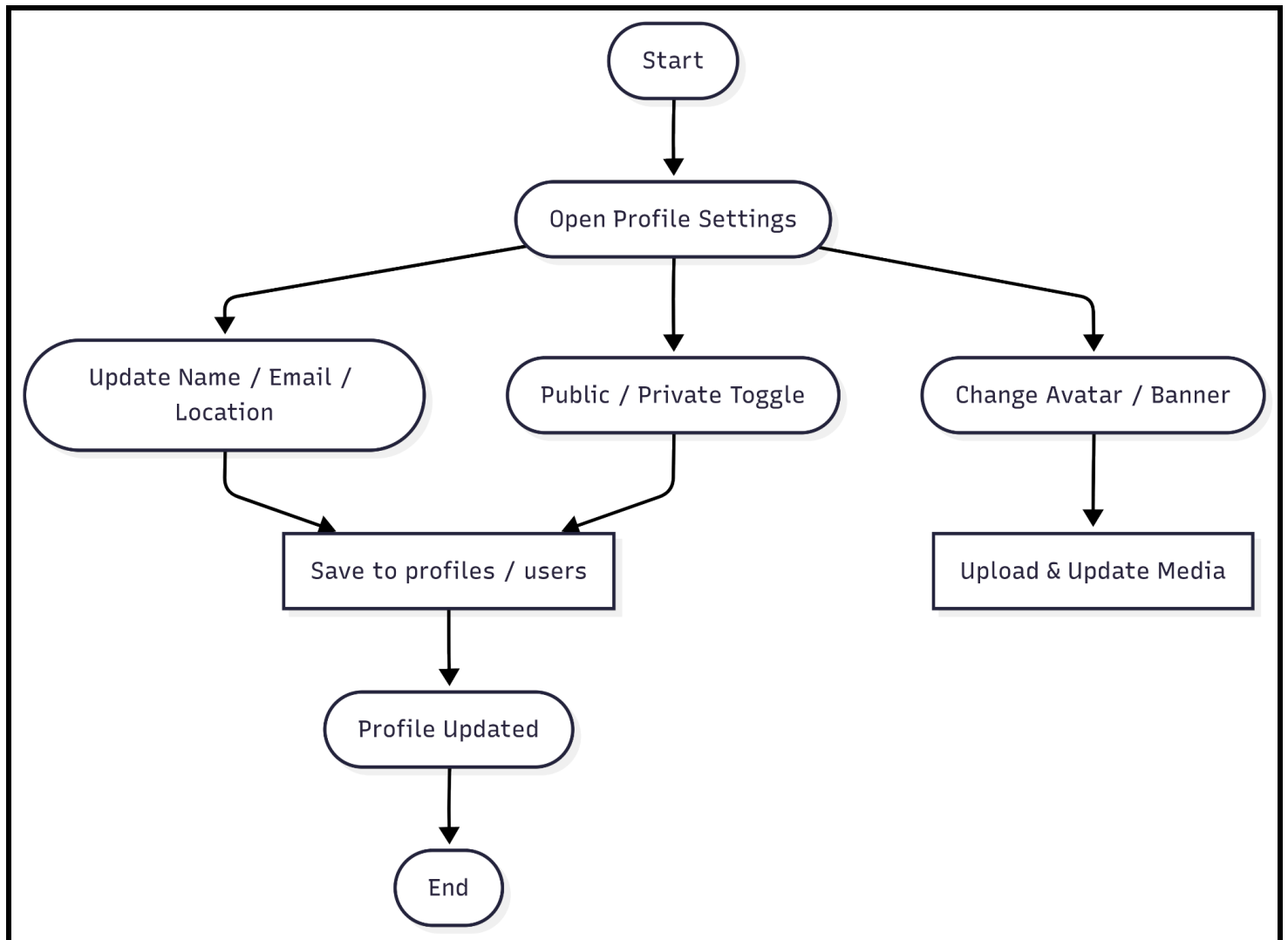


Fig. 2.2.14 System User Profile Flow

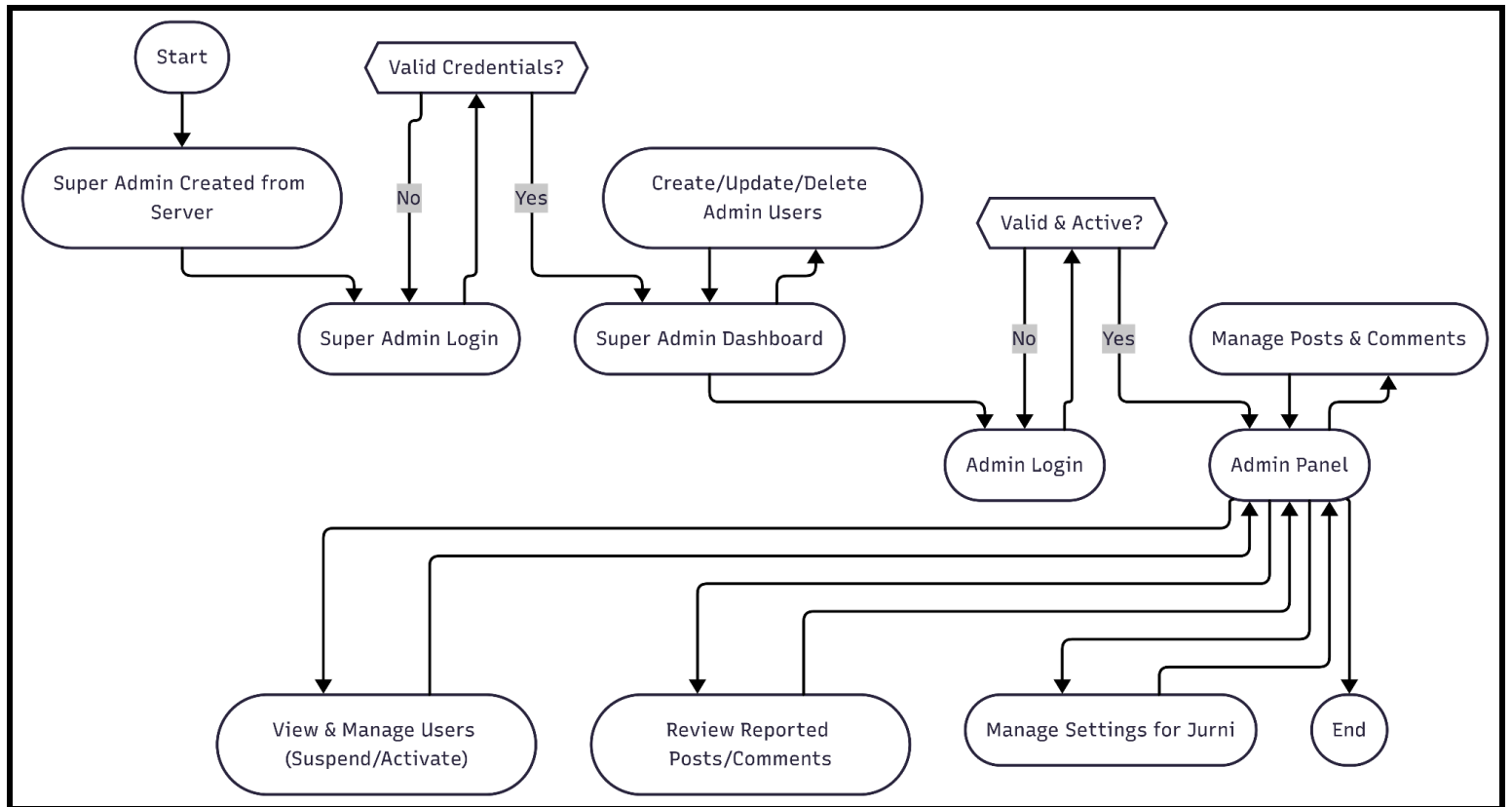


Fig. 2.2.15 System Admin Flow

2.3 Database Side

2.3.1 ERD (Entity Relationship Diagram)

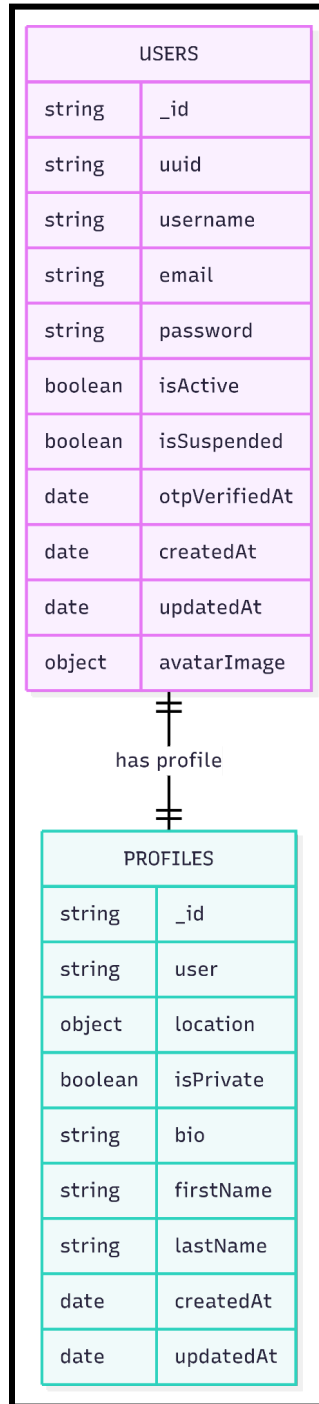


Fig. 2.3.1 ERD Authentication Core

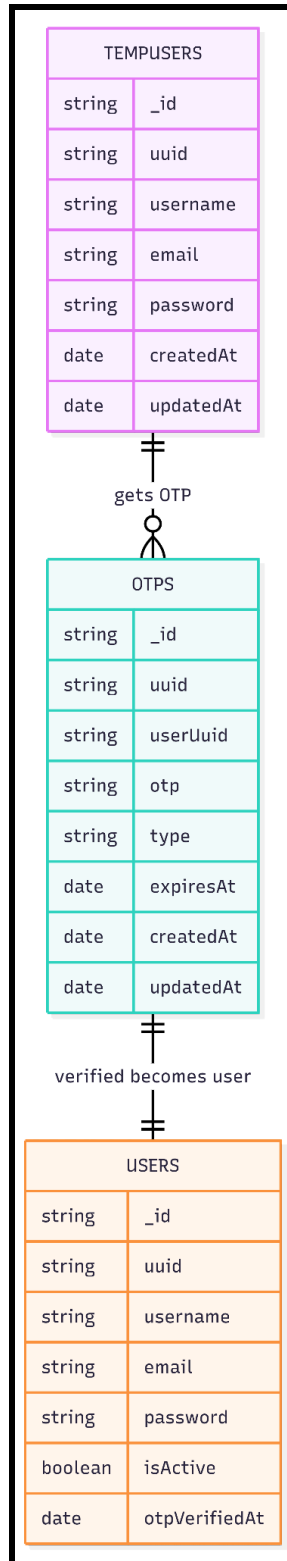


Fig. 2.3.2 ERD Registration and OTP Verification System

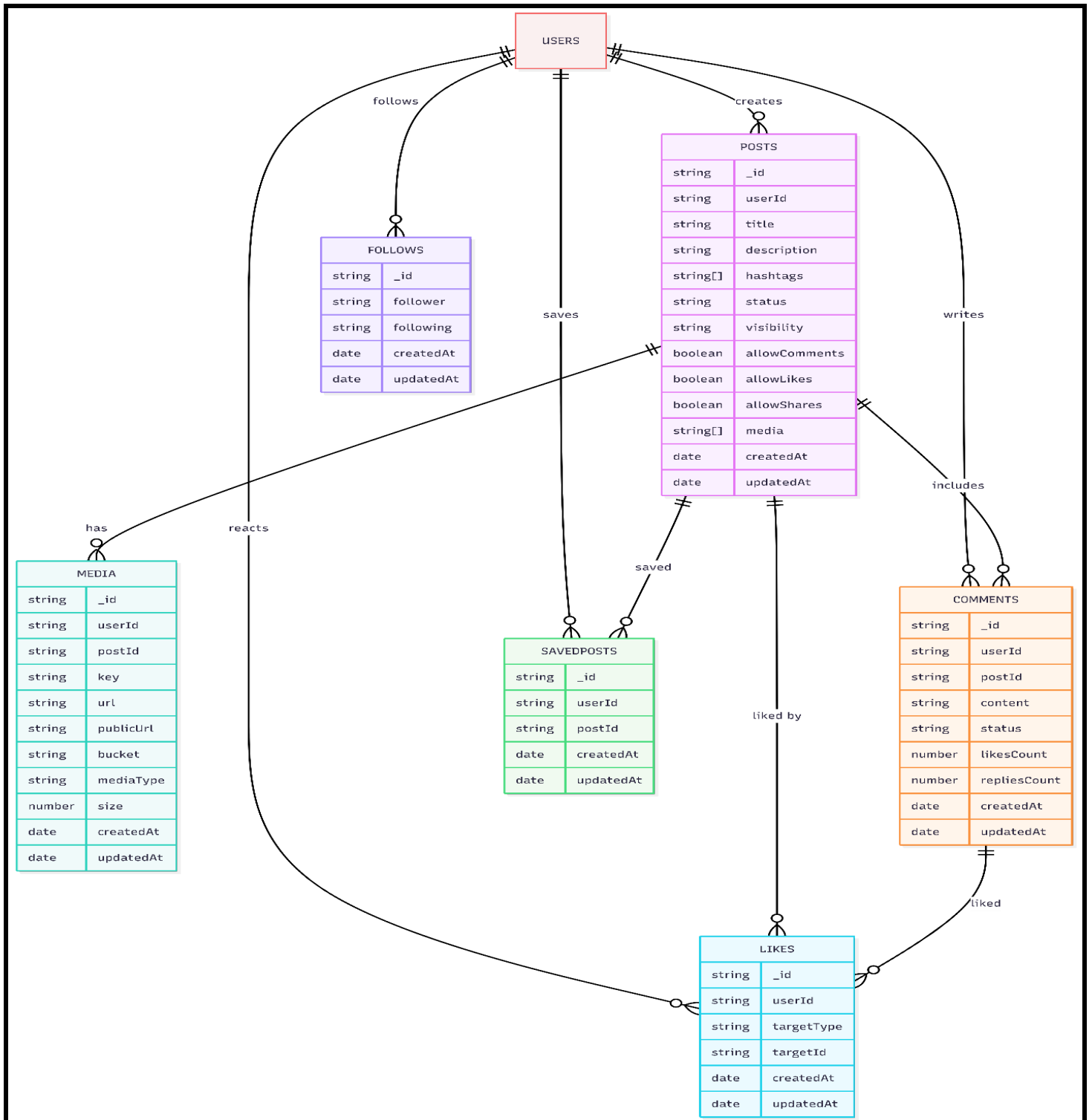


Fig. 2.3.3 ERD Posts and Social System

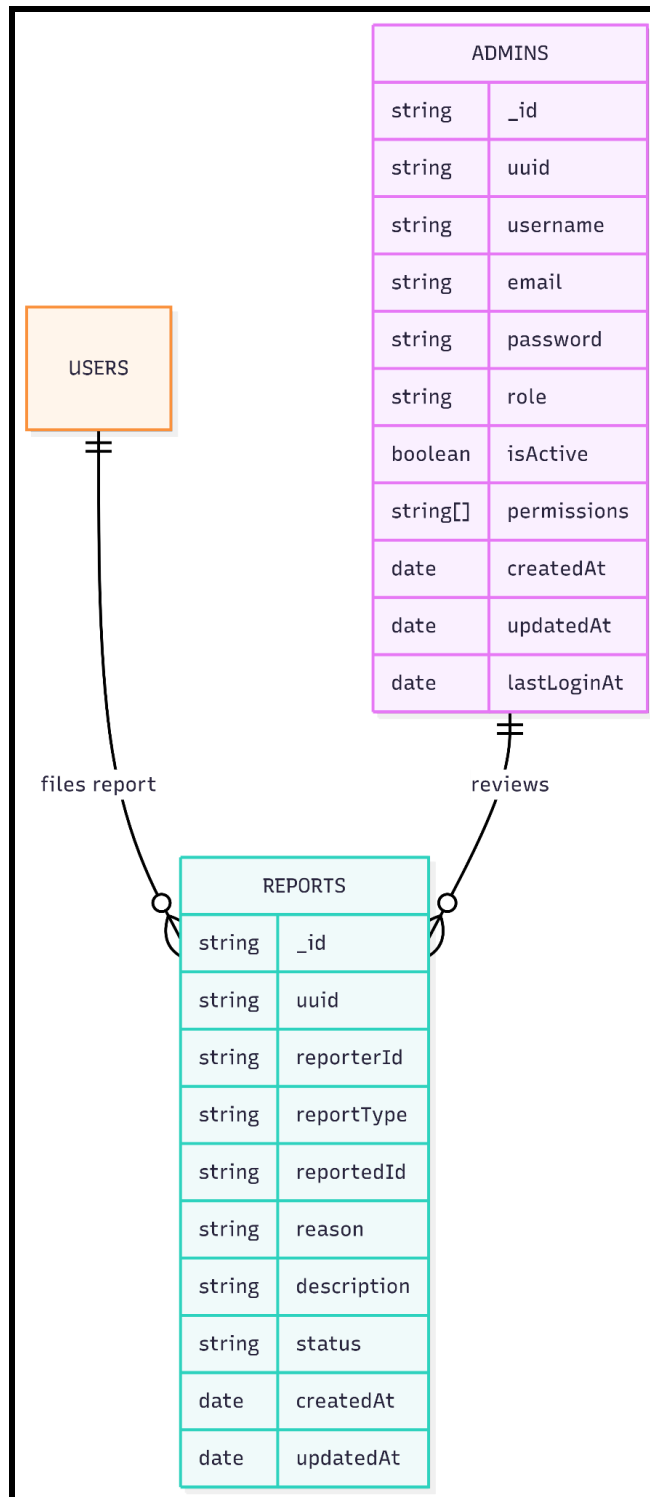


Fig. 2.3.4 ERD Admin and Reporting System

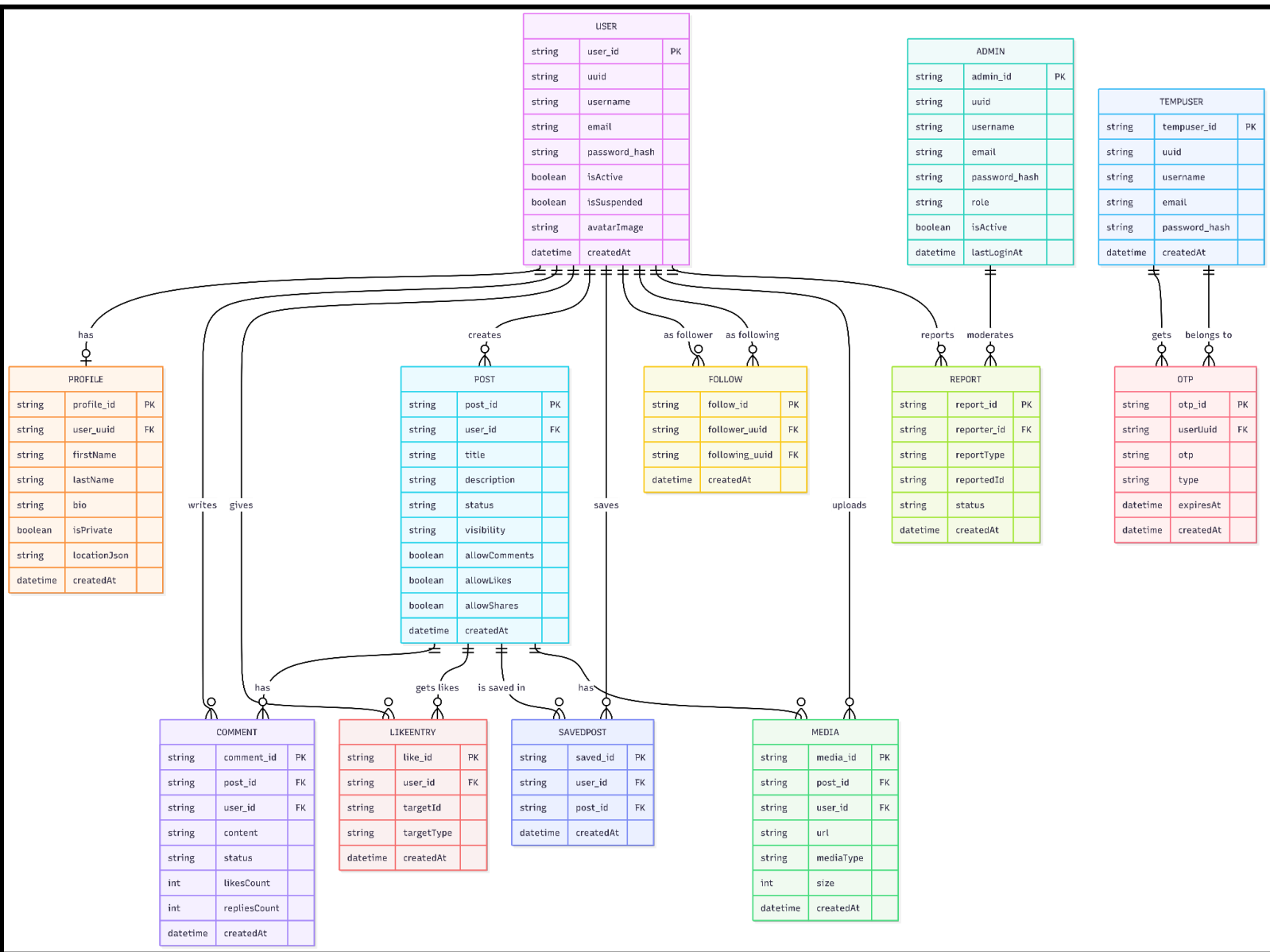
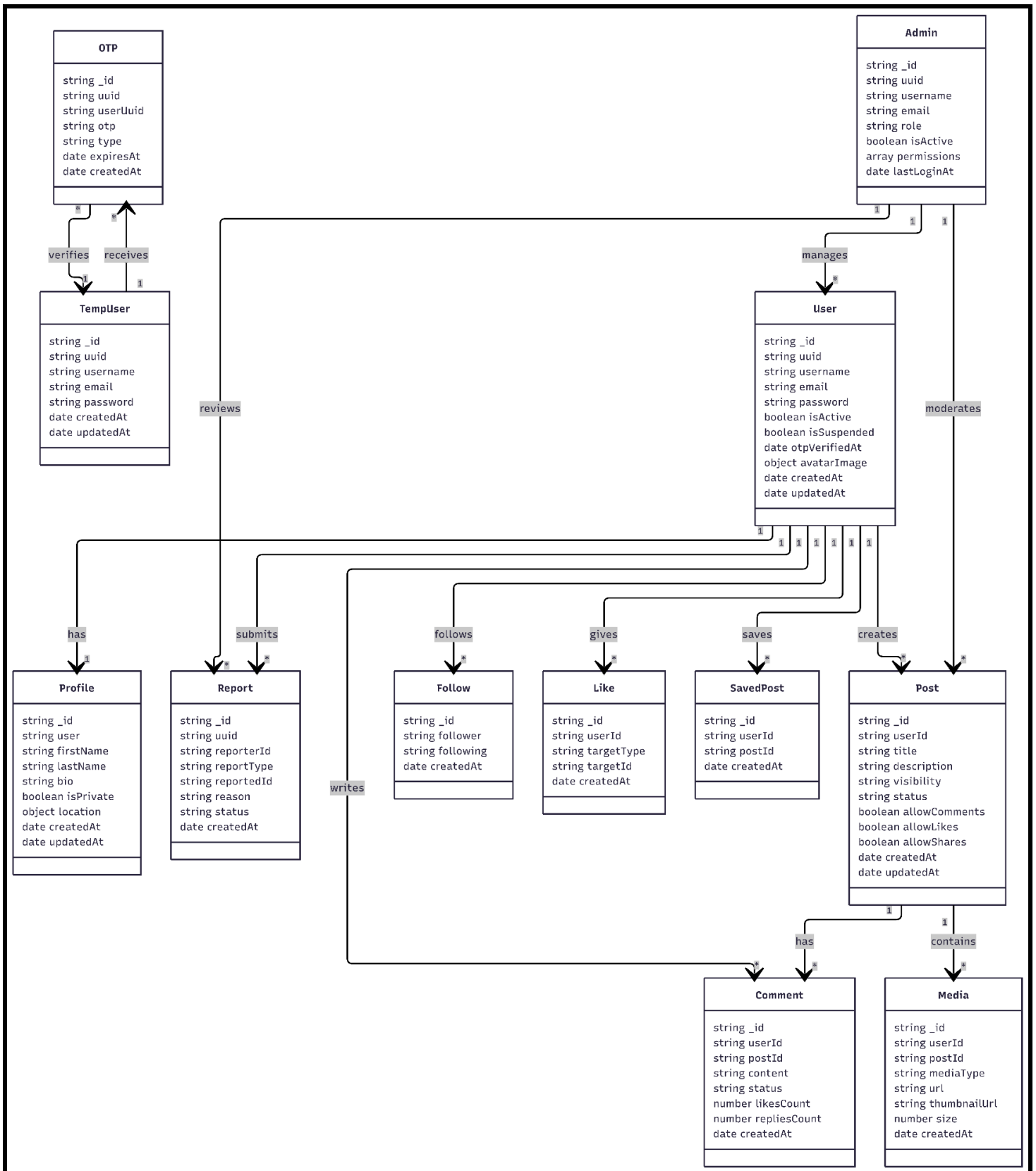
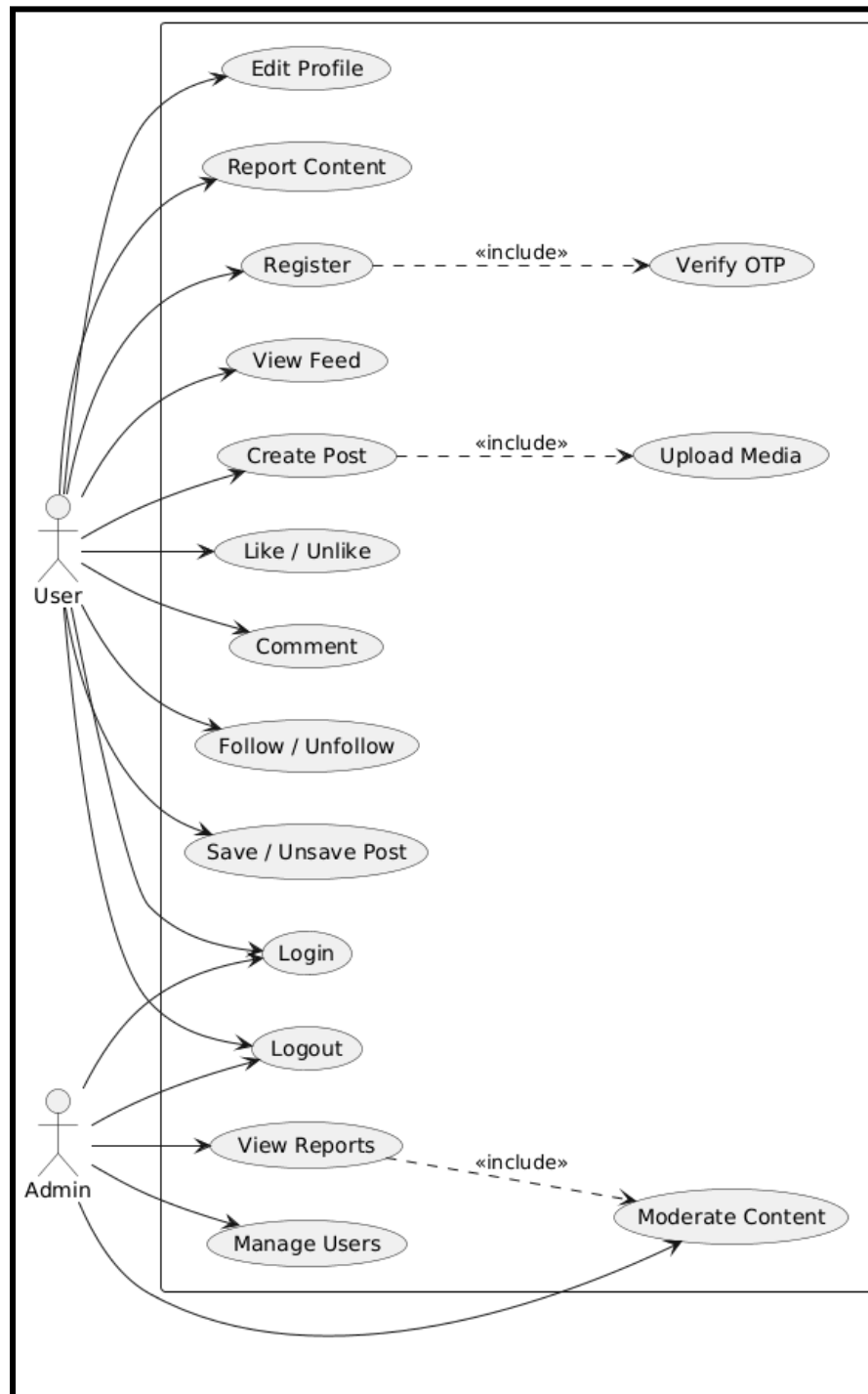


Fig. 2.3.5 ERD Full System

2.4 Class Diagram



2.5 Use Case Diagram



2.6 Sequence Diagram

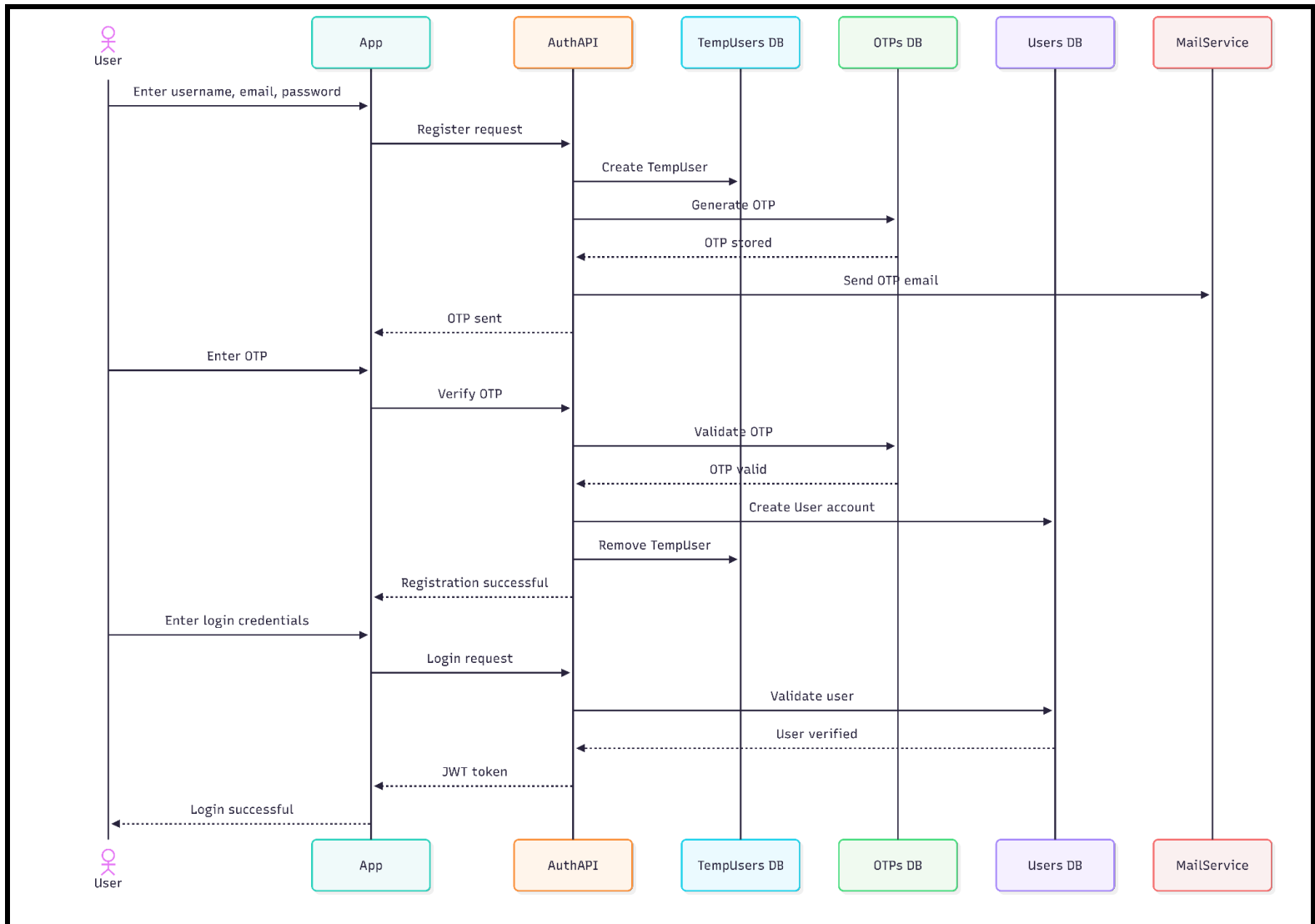


Fig. 2.6.1 Sequence Auth Diagram

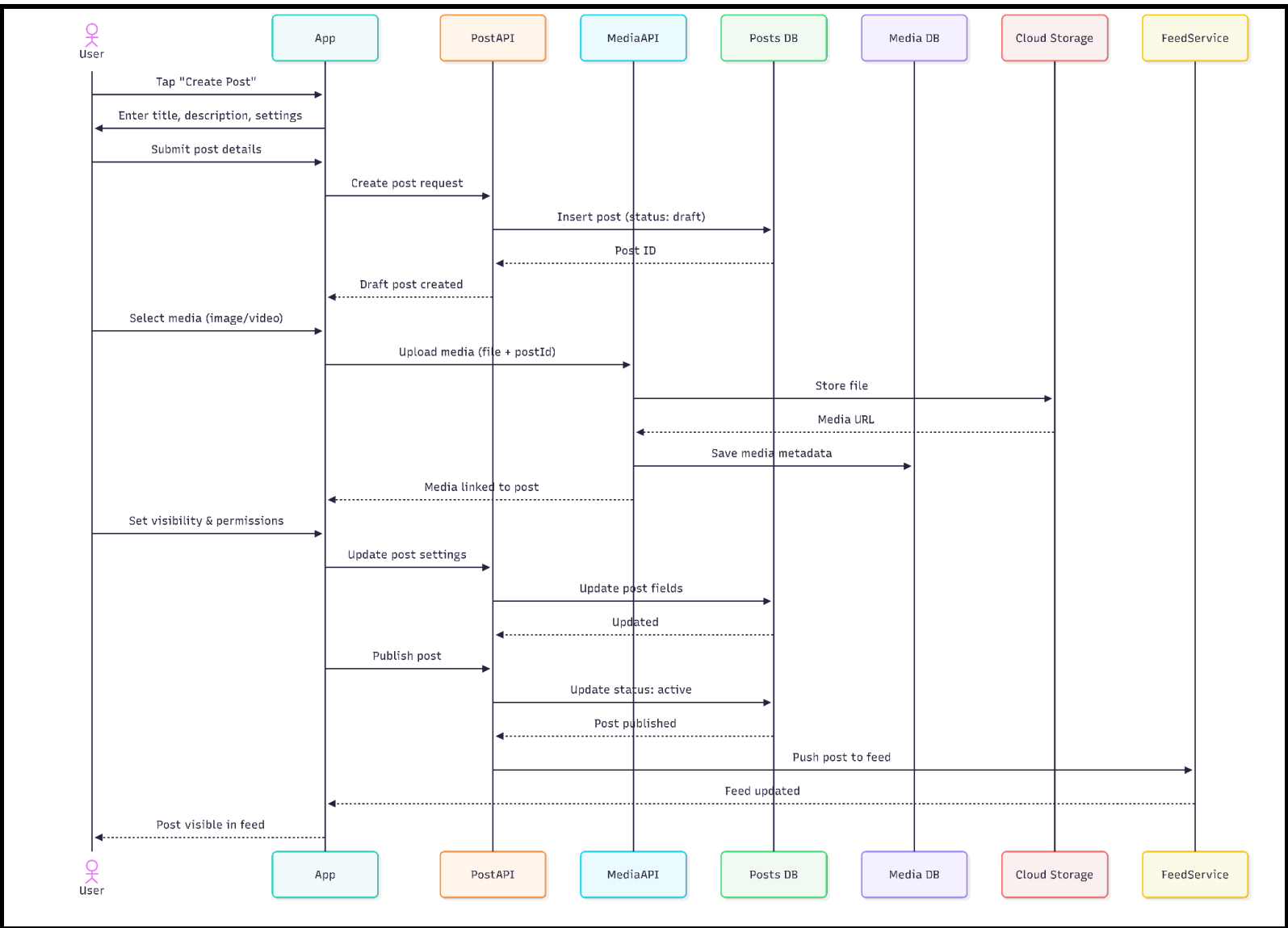


Fig. 2.6.2 Sequence Create Post with Media Upload

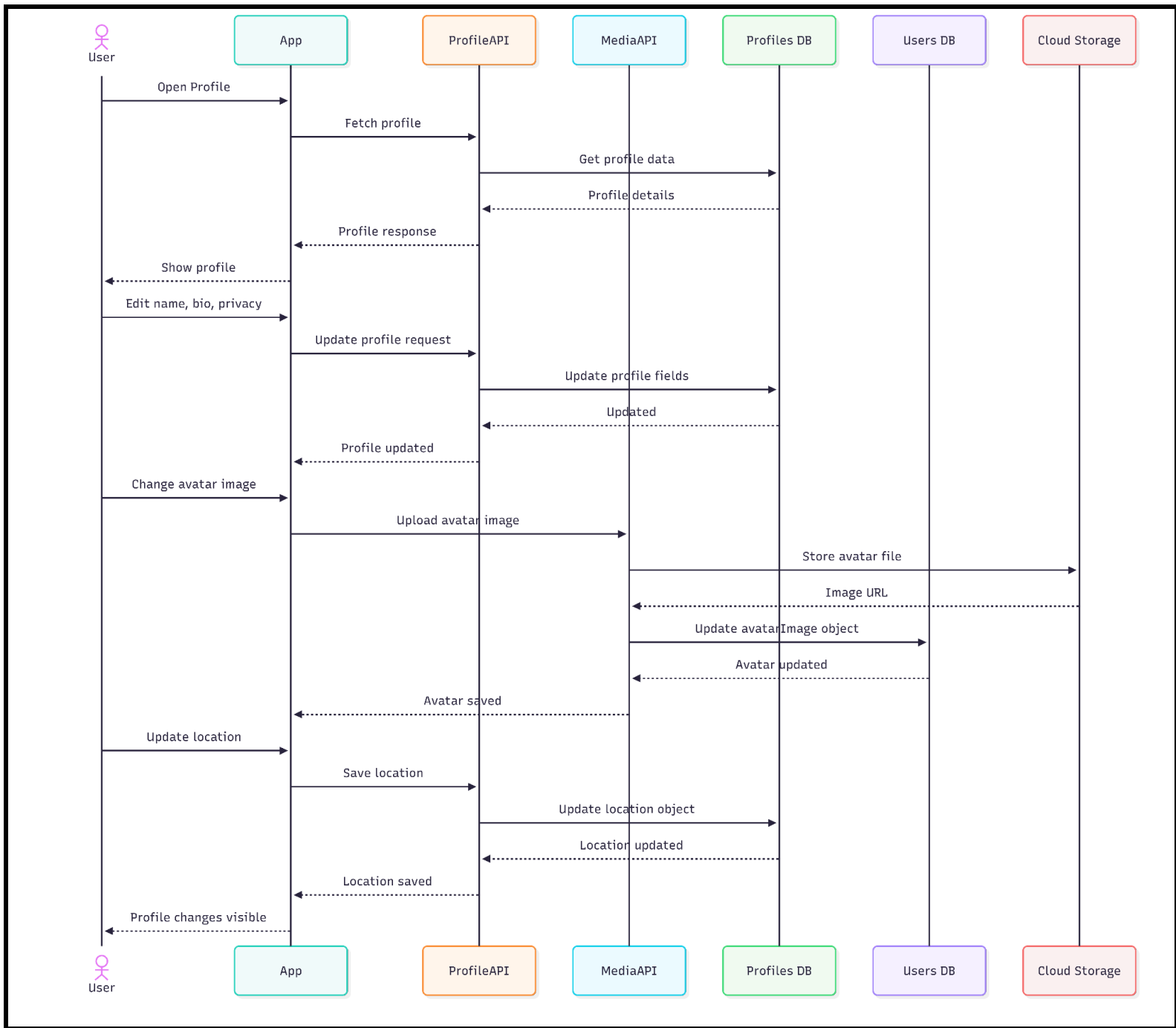


Fig. 2.6.3 Sequence User Profile View & Update

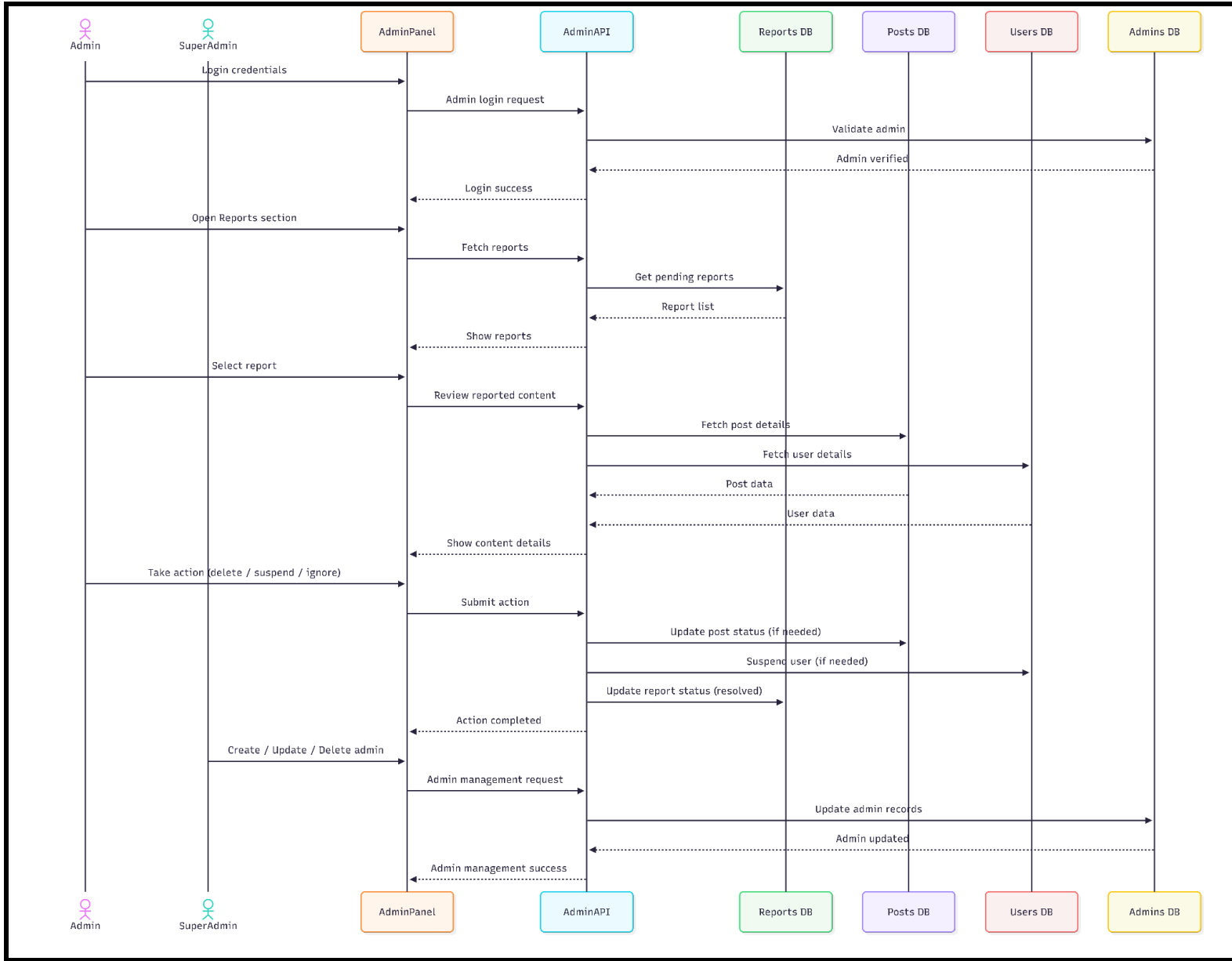


Fig. 2.6.4 Sequence Admin View & Update

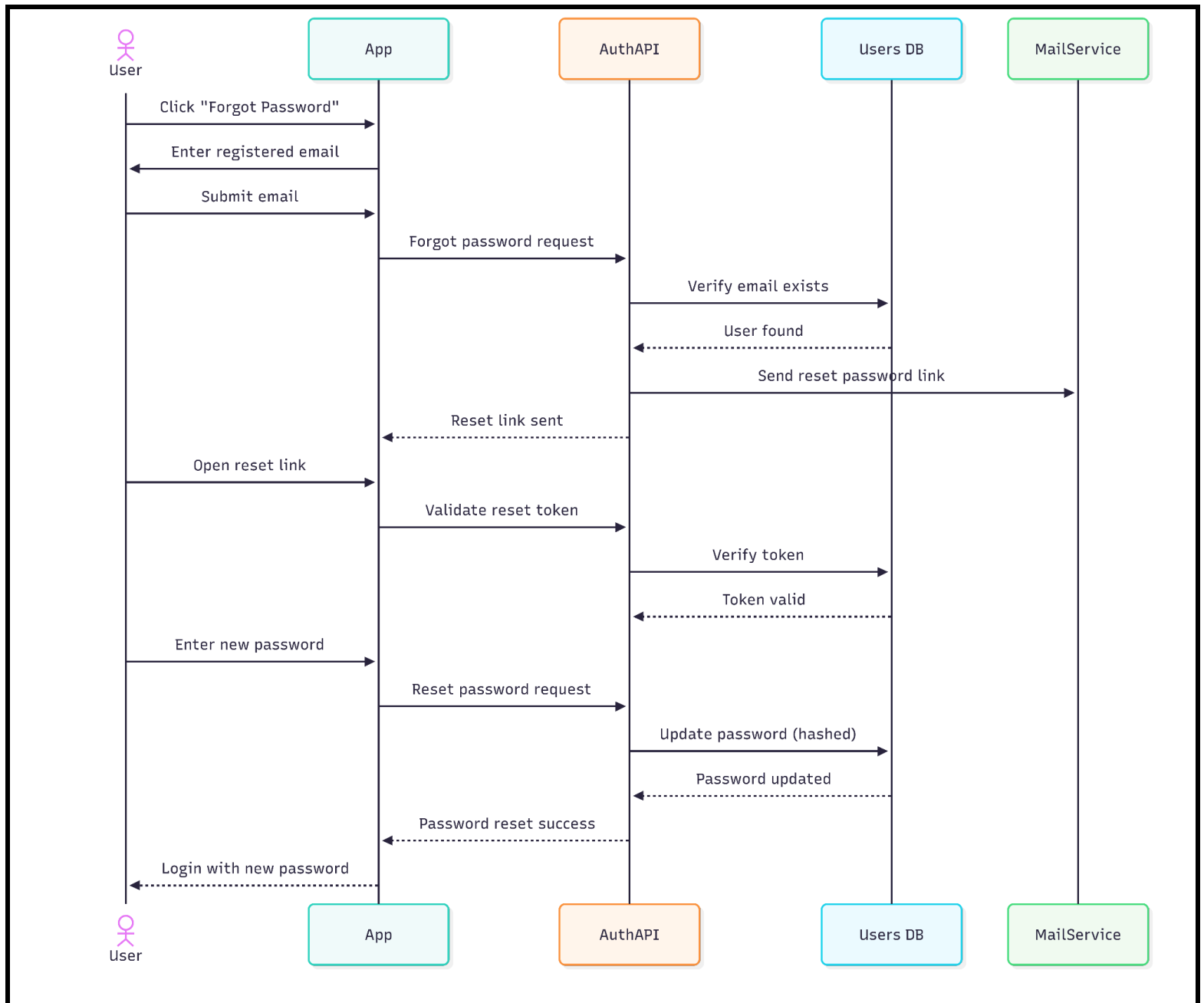


Fig. 2.6.5 Forgot Password & Reset Password

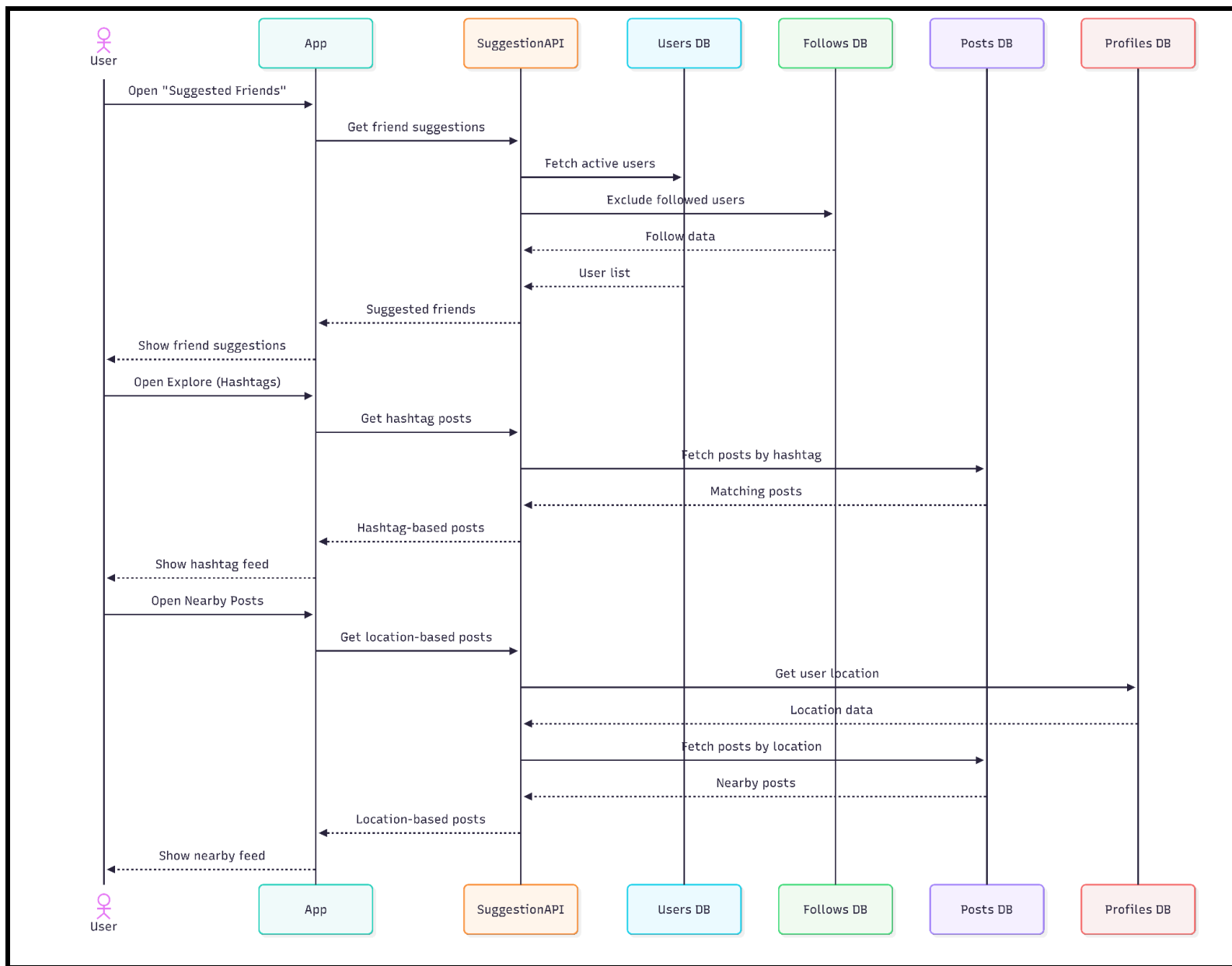


Fig. 2.6.6 Sequence Suggestions (Friends, Hashtags, Location)

3. Design

3.1 Data Dictionary

Field Name	Data Type	Description
_id	ObjectId	Temporary user ID.
uuid	String	Temp user's unique identifier.
username	String	Temporary username.
email	String	Temporary email.
password	String	Encrypted temporary password.
createdAt	Date	Temp user created time.
updatedAt	Date	Last updated time.
__v	Number	Version key.

Table. 3.1.1 Data Dictionary Tempusers

Field Name	Data Type	Description
_id	ObjectId	OTP record ID.
uuid	String	Unique OTP identifier.
userUuid	String	Temp user UUID linked to OTP.
otp	String	OTP code.
type	String	OTP type (registration, etc.).
expiresAt	Date	OTP expiry time.
createdAt	Date	OTP creation time.
updatedAt	Date	Last update time.
__v	Number	Version key.

Table. 3.1.2 Data Dictionary Otps

Field Name	Data Type	Description
_id	ObjectId	User ID.
uuid	String	User unique identifier.
username	String	Username.
email	String	Email address.
password	String	Encrypted password.
isActive	Boolean	User active or not.
otpVerifiedAt	Date	OTP verified time.
createdAt	Date	Account creation time.
updatedAt	Date	Last update time.
__v	Number	Version key.
avatarImage	Object	User profile image info.
→ key	String	Storage file path.
→ url	String	Image URL.
→ publicUrl	String	Public image link.
→ bucket	String	Storage bucket name.
→ size	Number	Image size.
→ contentType	String	Image type.
→ mediaId	String	Linked media ID.
isSuspended	Boolean	Account suspended or not.

Table. 3.1.3 Data Dictionary Users

Field Name	Data Type	Description
_id	ObjectId	Admin ID.
uuid	String	Admin unique identifier.
username	String	Admin username.
email	String	Admin email.
password	String	Encrypted password.
role	String	Admin role (super_admin/admin).
isActive	Boolean	Admin active or not.
permissions	Array	List of allowed actions.
createdAt	Date	Creation time.
updatedAt	Date	Last update time.
__v	Number	Version key.
lastLoginAt	Date	Last login time.

Table. 3.1.4 Data Dictionary Admins

Field Name	Data Type	Description
_id	ObjectId	Profile ID.
user	String (UUID)	User UUID linked to profile.
firstName	String	User first name.
lastName	String	User last name.
bio	String	Profile bio.
location	Object	Location information.
isPrivate	Boolean	Profile private or not.
createdAt	Date	Profile created time.
updatedAt	Date	Last update time.
__v	Number	Version key.

Table. 3.1.5 Data Dictionary Profiles

Field Name	Data Type	Description
_id	ObjectId	Post ID.
userId	ObjectId	ID of user who created the post.
title	String	Post title.
description	String	Post content/description.
hashtags	Array	List of hashtags.
status	String	Post status (active/deleted).
visibility	String	public or private.
allowComments	Boolean	Comments allowed or not.
allowLikes	Boolean	Likes allowed or not.
allowShares	Boolean	Shares allowed or not.
media	Array(ObjectId)	Media files linked to post.
createdAt	Date	Post creation time.
updatedAt	Date	Last update time.
__v	Number	Version key.

Table. 3.1.6 Data Dictionary Posts

Field Name	Data Type	Description
_id	ObjectId	Media ID.
userId	ObjectId	User who uploaded media.
postId	ObjectId	Post linked to the media.
key	String	Storage file path.
url	String	Main media URL.
publicUrl	String	Public viewing URL.
bucket	String	Cloud storage bucket name.
mediaType	String	Media type (image/video).
thumbnailUrl	String	Thumbnail link.
size	Number	File size in bytes.
createdAt	Date	Upload time.
updatedAt	Date	Last update.
__v	Number	Version key.

Table. 3.1.7 Data Dictionary Media

Field Name	Data Type	Description
_id	ObjectId	Saved post ID.
userId	ObjectId	User who saved the post.
postId	ObjectId	Post that is saved.
createdAt	Date	Saved time.
updatedAt	Date	Last update time.
__v	Number	Version key.

Table. 3.1.8 Data Dictionary Saveposts

Field Name	Data Type	Description
_id	ObjectId	Like record ID.
userId	ObjectId	Users who liked it.
targetType	String	Type (post/comment).
targetId	ObjectId	ID of liked post/comment.
createdAt	Date	Like time.
updatedAt	Date	Last update.
__v	Number	Version key.

Table. 3.1.9 Data Dictionary Likes

Field Name	Data Type	Description
_id	ObjectId	Comment ID.
userId	ObjectId	User who commented.
postId	ObjectId	Post on which comment is made.
content	String	Comment text.
status	String	active or deleted.
likes	Array	Users who liked comment.
likesCount	Number	Total likes.
repliesCount	Number	Total replies.
createdAt	Date	Comment time.
updatedAt	Date	Last update.
__v	Number	Version key.

Table. 3.1.10 Data Dictionary Comment

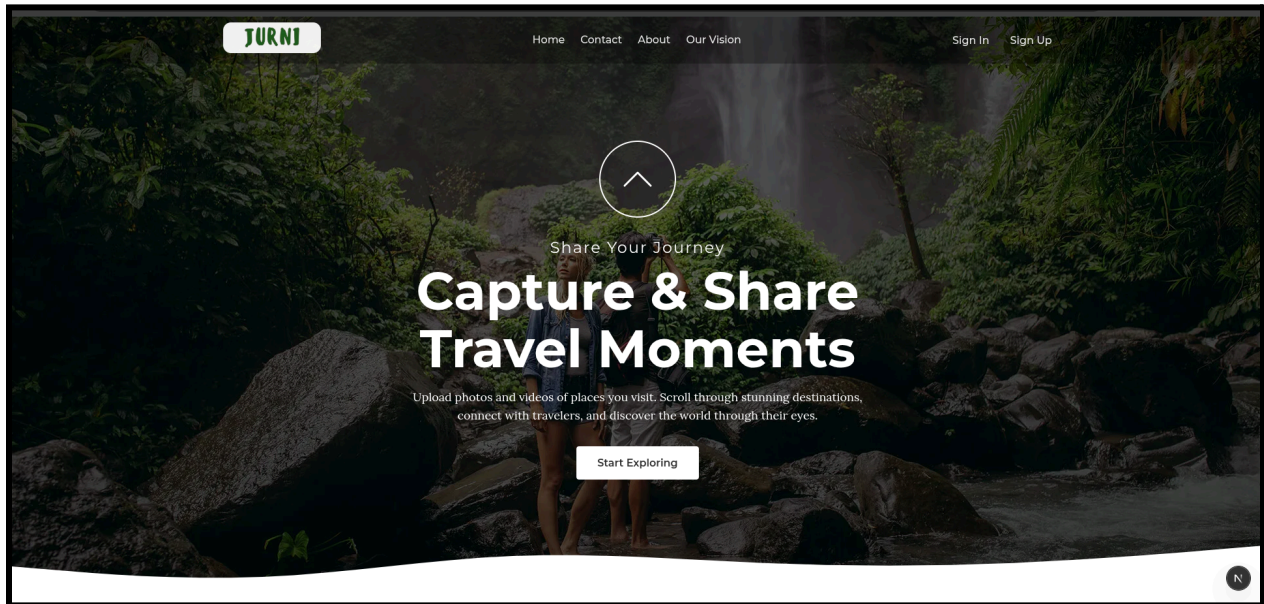
Field Name	Data Type	Description
_id	ObjectId	Report ID.
uuid	String	Unique report identifier.
reporterId	ObjectId	User who reported.
reportType	String	Type (post/user/comment).
reportedId	String/ObjectId	ID of reported item.
reason	String	Reason for reporting.
description	String	Additional details.
status	String	pending or resolved.
createdAt	Date	Report time.
updatedAt	Date	Last update.
__v	Number	Version key.

Table. 3.1.11 Data Dictionary Reports

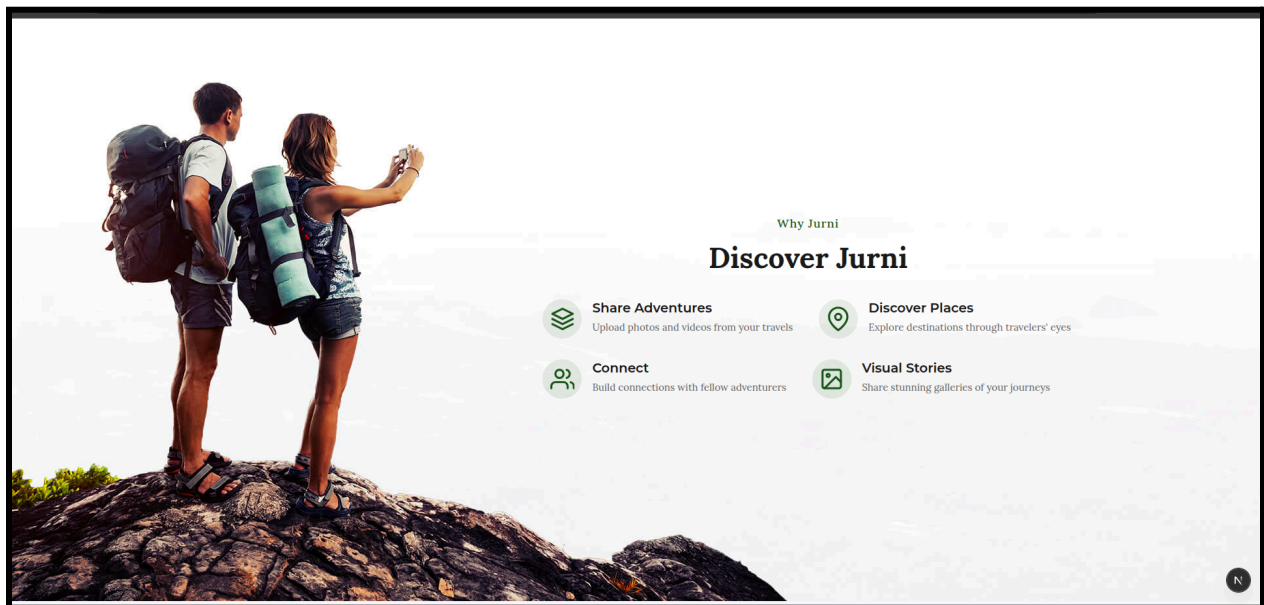
Field Name	Data Type	Description
_id	ObjectId	Follow record ID.
follower	String (UUID)	User who follows.
following	String (UUID)	User being followed.
createdAt	Date	Follow time.
updatedAt	Date	Last update.
__v	Number	Version key.

Table. 3.1.12 Data Dictionary Follows

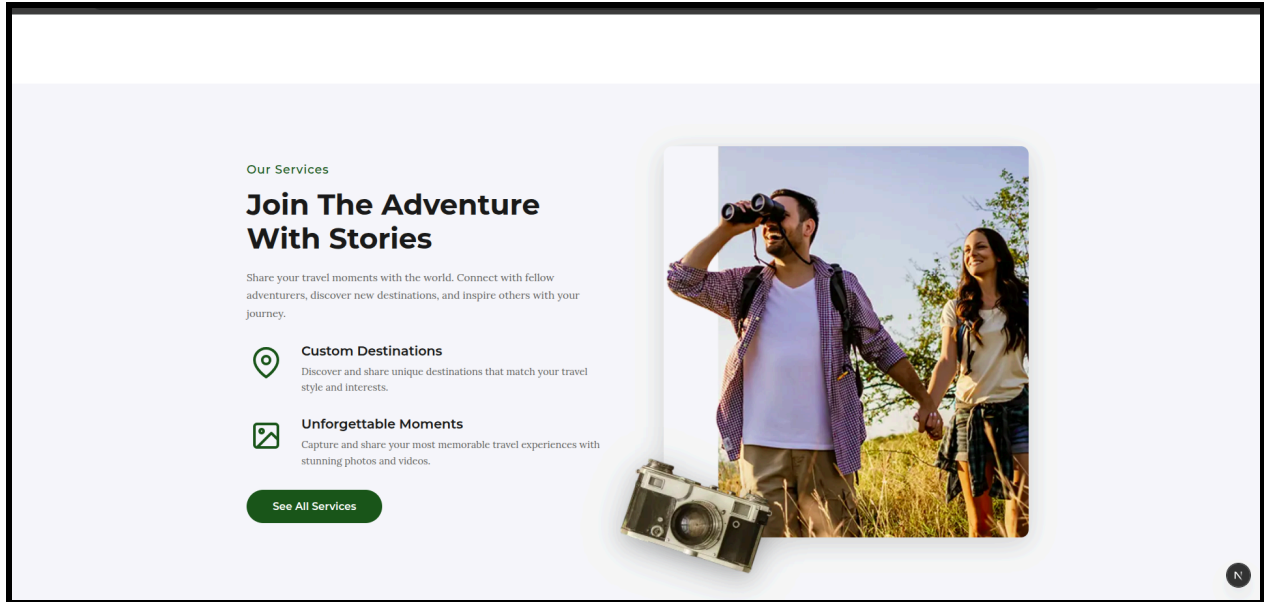
3.2 Screens / Interfaces



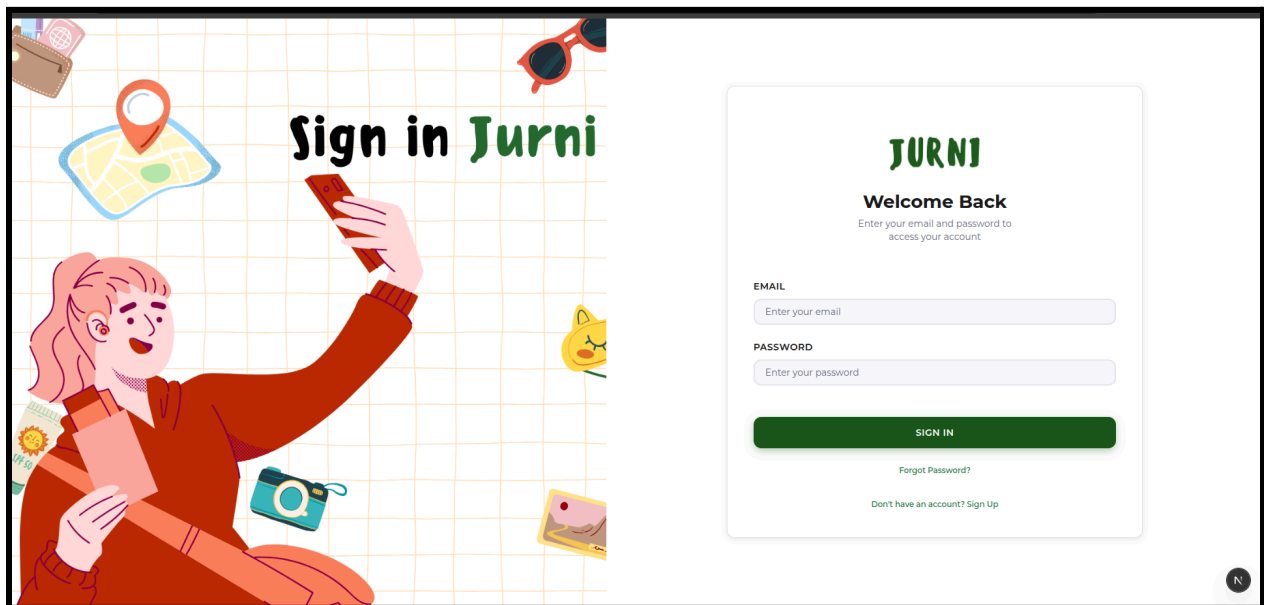
Screen 3.2.1 Website 1



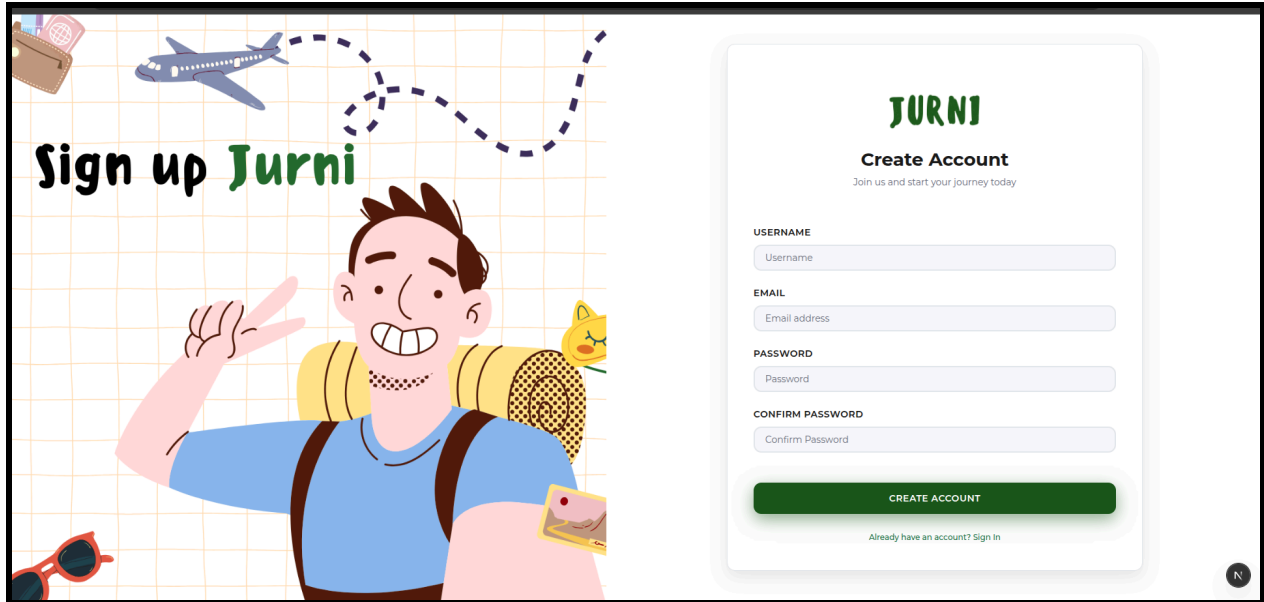
Screen 3.2.2 Website 2



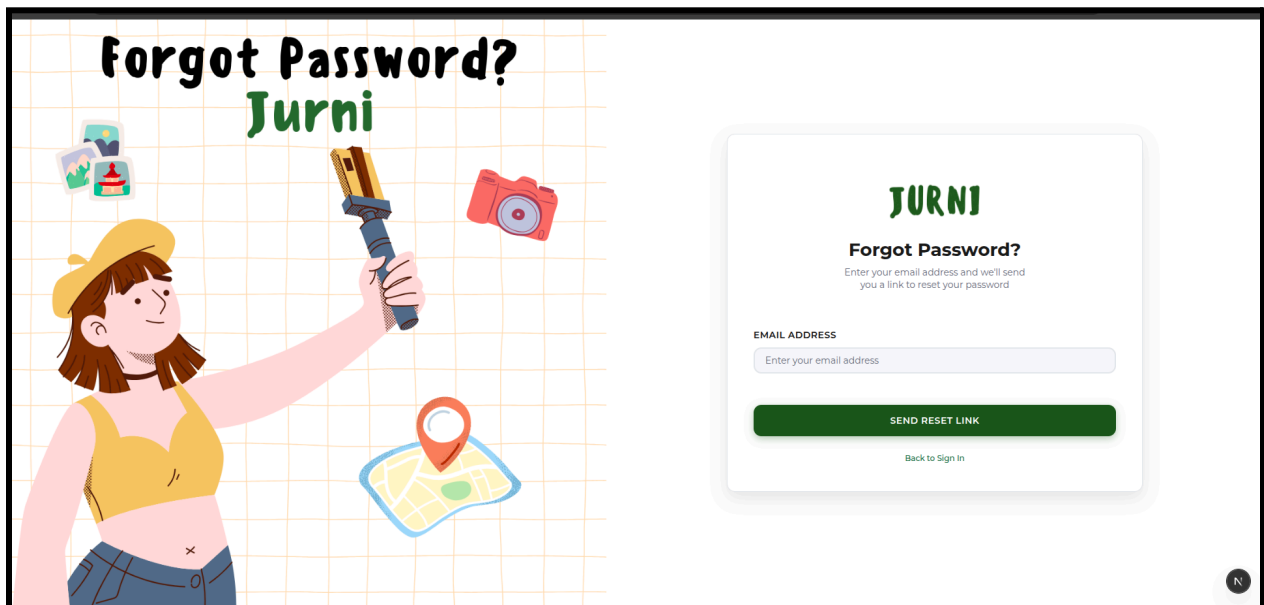
Screen 3.2.3 Website 3



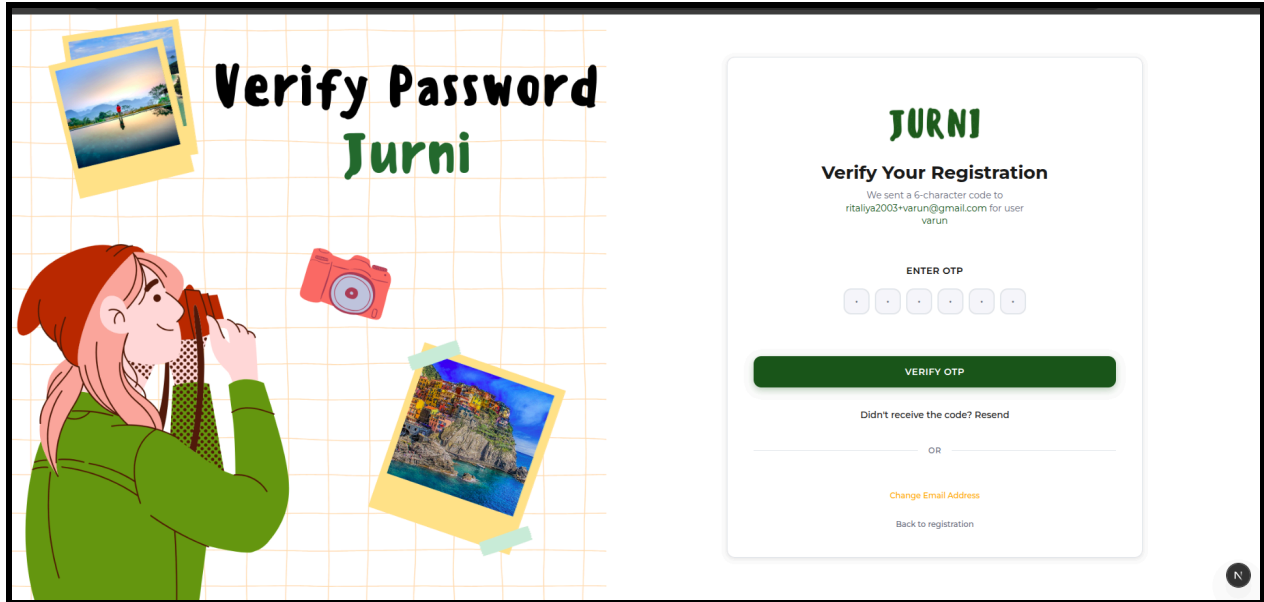
Screen 3.2.4 Sign in Screen



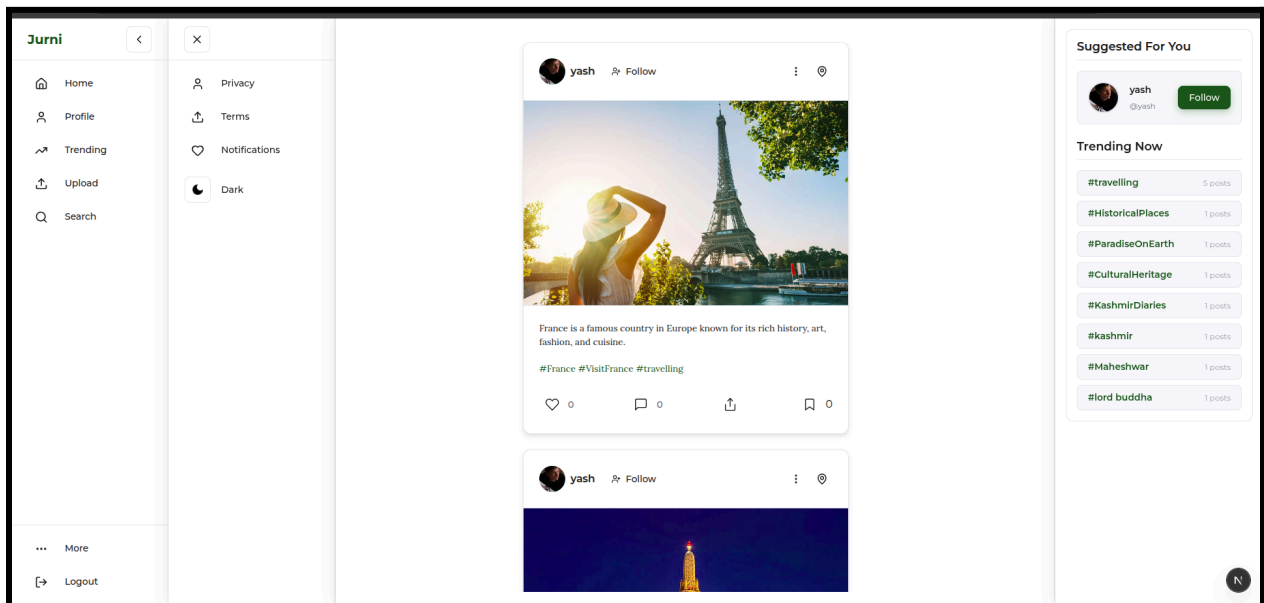
Screen 3.2.5 Sign up Screen



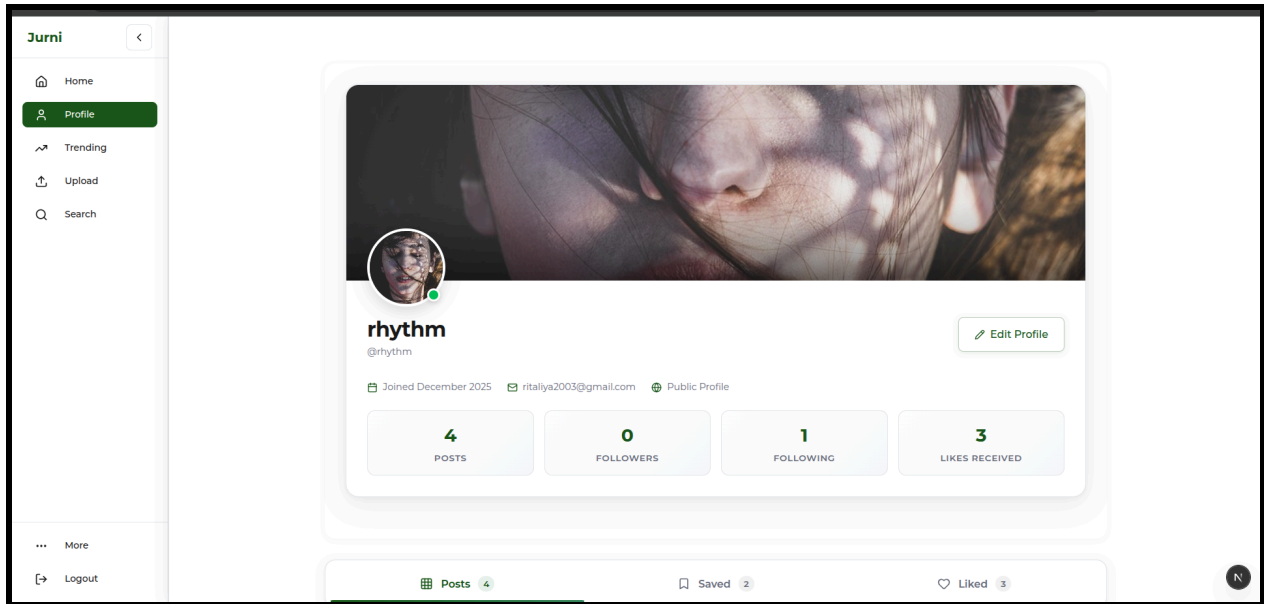
Screen 3.2.6 Forgot Password Screen



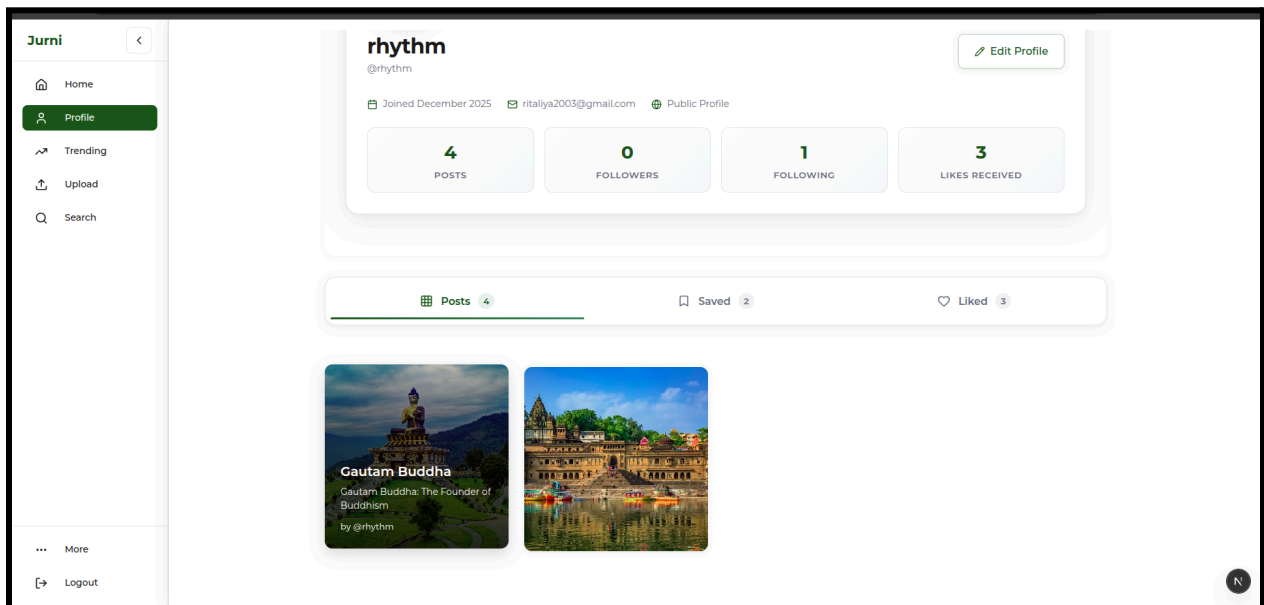
Screen 3.2.7 Verify Password Screen



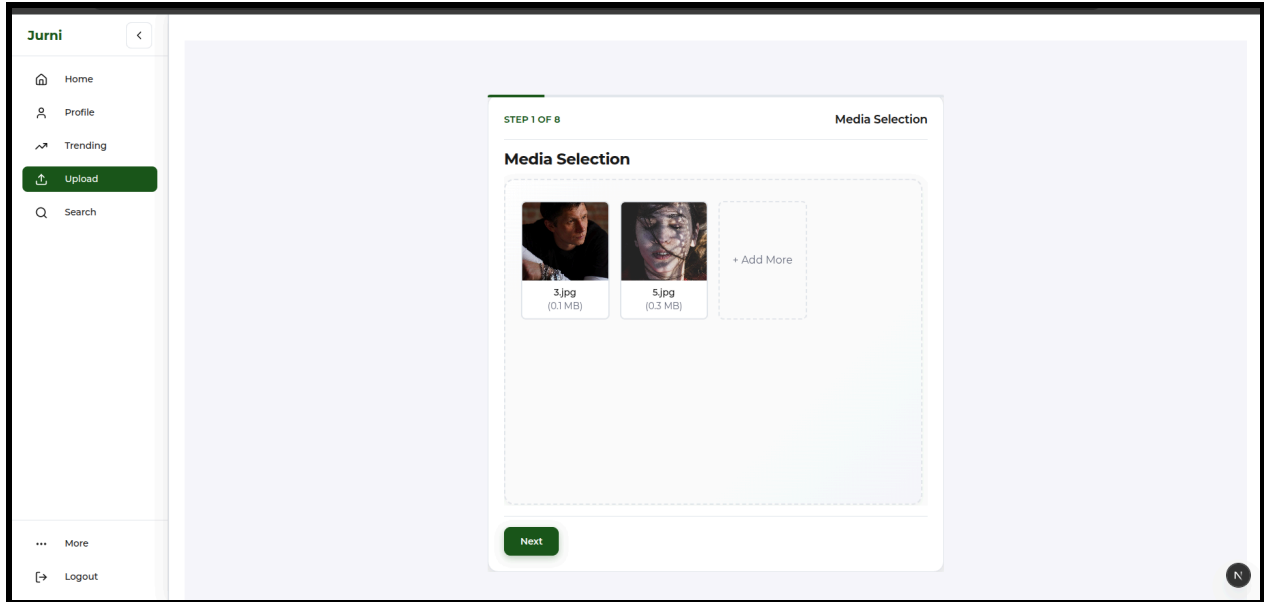
Screen 3.2.8 Home Screen



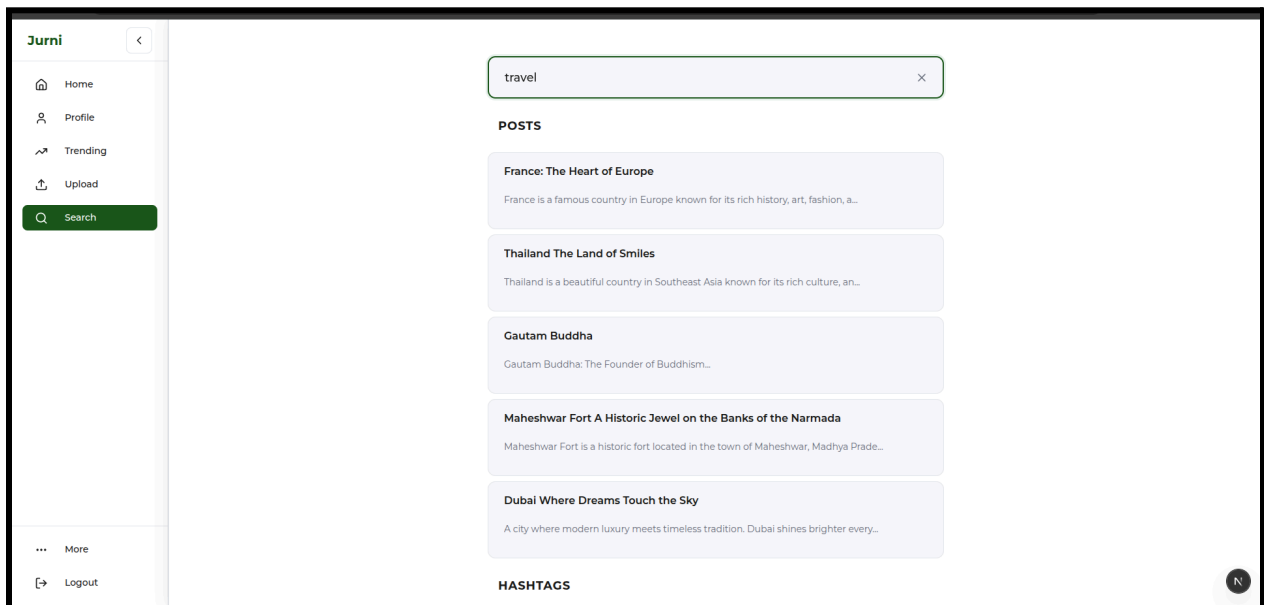
Screen 3.2.9 Profile Screen



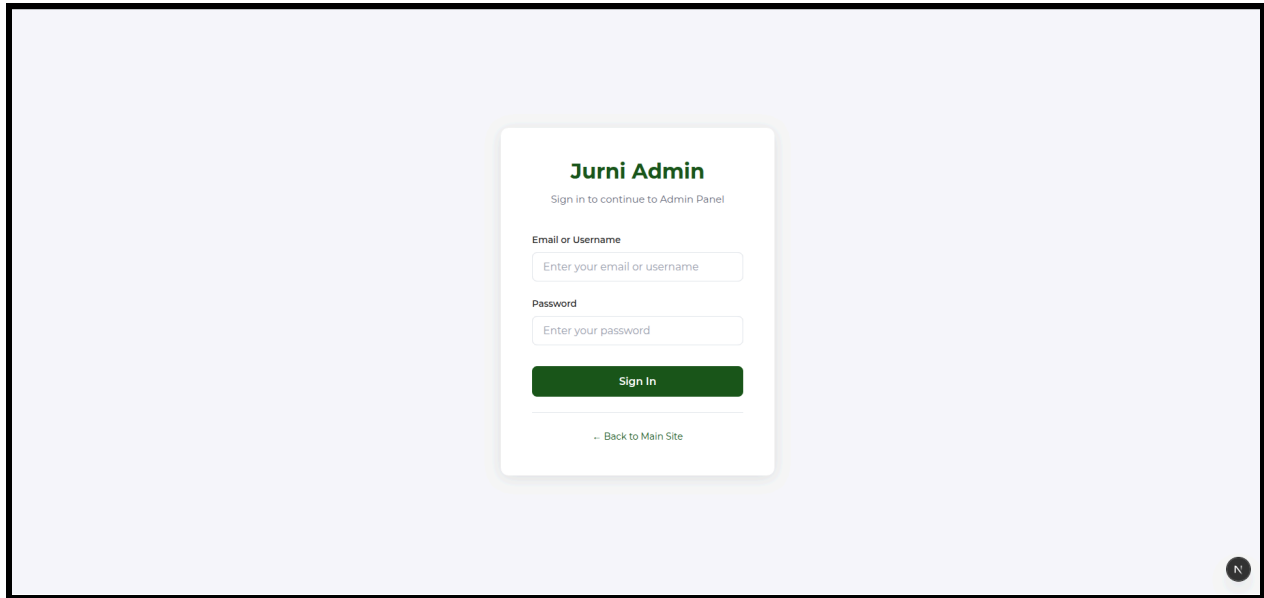
Screen 3.2.10 Profile Screen (Posts)



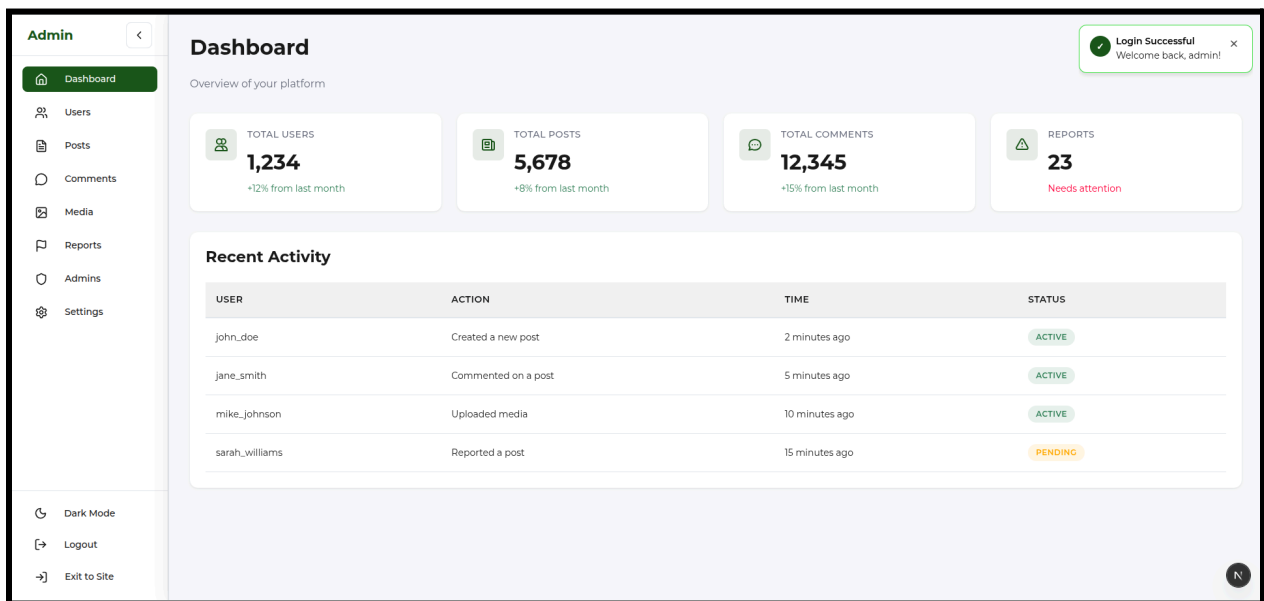
Screen 3.2.11 Upload Post Scree



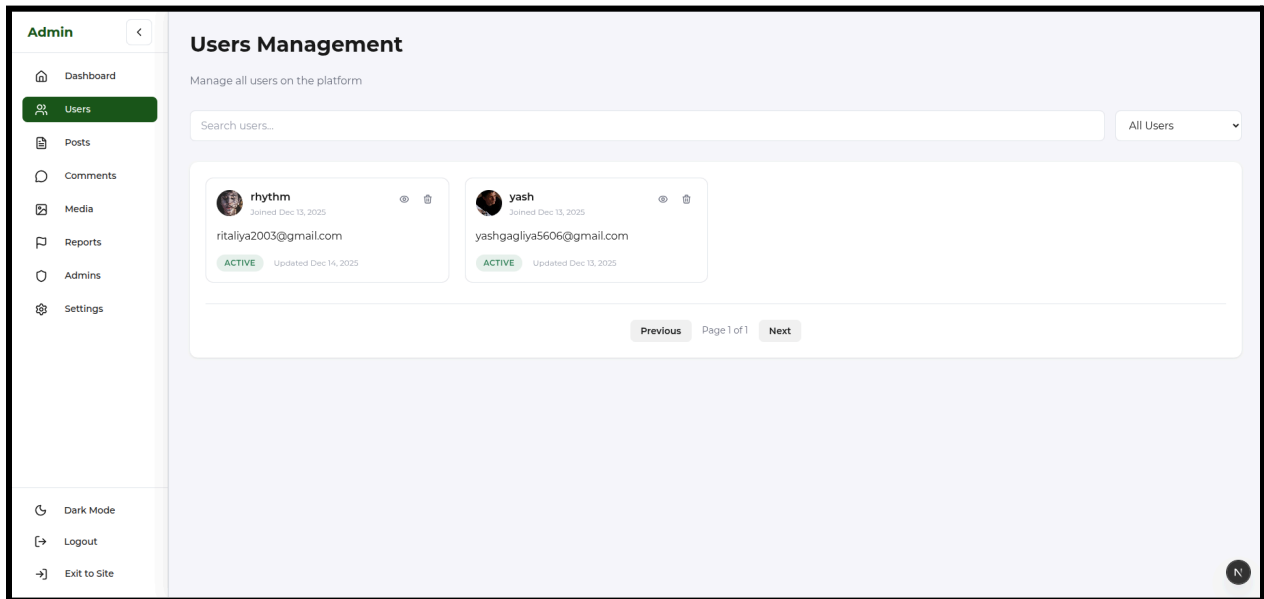
Screen 3.2.12 Search Screen



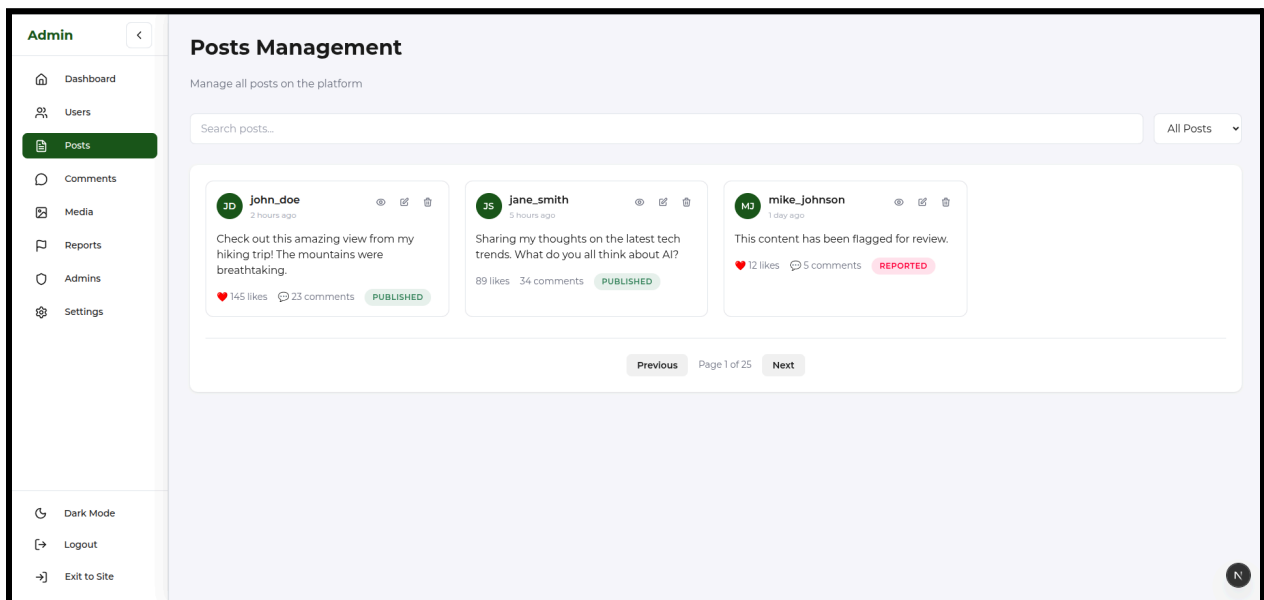
Screen 3.2.13 Admin login Screen



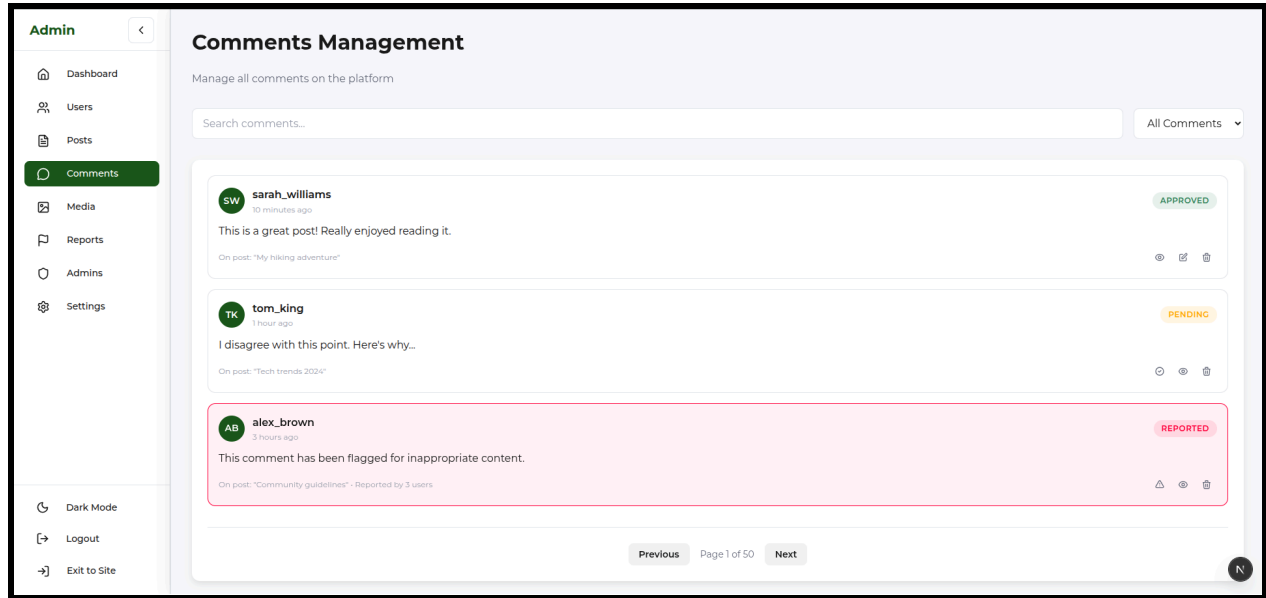
Screen 3.2.14 Admin Dashboard Screen



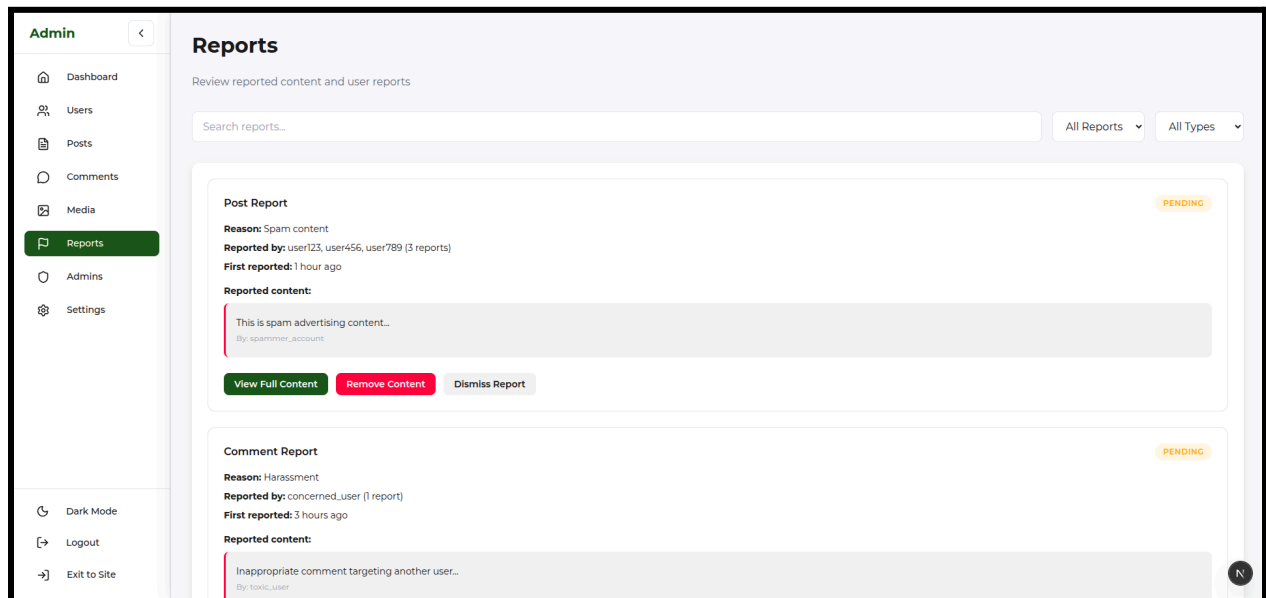
Screen 3.2.15 User Dashboard Screen



Screen 3.2.16 Post Dashboard Screen



Screen 3.2.17 Comments Dashboard Screen



Screen 3.2.18 Reports Dashboard Screen

4. Testing

4.1 White Box Testing

4.1.1 Authentication System

Tables : OTP

TC ID	Input	Internal Logic	Expected
JURNI-WB-001	OTP = correct	otp == storedOtp AND notExpired	User created in USERS
JURNI-WB-002	OTP = wrong	otp != storedOtp	Verification failed
JURNI-WB-003	OTP expired	now > expiresAt	Reject OTP
JURNI-WB-004	TempUser exists	tempUser.deleted=false	Convert temp to real user

Tables : USERS

TC ID	Input	Logic	Expected
JURNI-WB-005	correct pass	bcrypt.compare() → true	JWT created
JURNI-WB-006	wrong pass	bcrypt.compare() → false	Login error
JURNI-WB-007	suspended user	isSuspended == true	Access denied

4.1.2 Profile System

Tables : PROFILES, USERS

TC ID	Input	Logic	Expected
JURNI-WB-010	Update profile	db.update() → valid fields only	Saved
JURNI-WB-011	Invalid location	JSON validator fails	Error

4.1.3 Post System

Tables : POSTS, MEDIA

TC ID	Input	Logic	Expected
JURNI-WB-020	Create post	title != null	Save post
JURNI-WB-021	No title	title == null	Validation error
JURNI-WB-022	Upload file	file.mime in allowed list	Media saved

4.1.4 Comment System

Tables : COMMENTS

TC ID	Internal Logic	Expected
JURNI-WB-030	comment != ""	Save comment
JURNI-WB-031	comment = ""	Reject

4.1.5 Likes System

Tables : LIKES

TC ID	Logic	Expected
JURNI-WB-040	check existing like	prevent duplicate

4.1.6 Follow System

Tables : FOLLOWS

TC ID	Logic	Expected
JURNI-WB-050	existing follow?	Prevent double follow

4.1.7 Report System

Tables: REPORTS, USERS, POSTS

TC ID	Logic	Expected
JURNI-WB-060	reportType valid?	Save
JURNI-WB-061	duplicate report	Reject

4.1.8 Admin Panel

Tables : ADMINS, REPORTS

TC ID	Logic	Expected
JURNI-WB-070	Check permissions[]	Allow/deny action
JURNI-WB-071	Deactivate user	user.isSuspended = true

4.2 Unit Testing/Validation Testing

ID	Function	Input	Expected
JURNI-UT-001	validateEmail()	abc@gmail.com	Valid
JURNI-UT-002	validateEmail()	abc@gmail	Invalid
JURNI-UT-003	hashPassword()	"12345"	bcrypt hash
JURNI-UT-004	checkFileType()	image/jpeg	Allowed
JURNI-UT-005	checkFileSize()	20MB	Reject

4.3 Black Box Testing

4.3.1 Registration

ID	Input	Expected
JURNI-BB-001	Valid details	OTP sent
JURNI-BB-002	Existing email	Error
JURNI-BB-003	Weak password	Error

4.3.2 Login

ID	Input	Expected
JURNI-BB-004	Correct	Login success
JURNI-BB-005	Wrong password	Error

4.3.3 Post System

ID	Input	Expected
JURNI-BB-010	Upload media	Post created
JURNI-BB-011	Large file	Error
JURNI-BB-012	Without title	Reject

4.3.4 Comments

ID	Input	Expected
JURNI-BB-020	Comment text	Saved
JURNI-BB-021	Empty	Error

4.3.5 Likes

ID	Input	Expected
JURNI-BB-030	Like once	Saved
JURNI-BB-031	Like twice	Prevent duplicate

4.3.6 Follow

ID	Input	Expected
JURNI-BB-040	Follow	Added
JURNI-BB-041	Follow again	Error

4.3.7 Reports

ID	Input	Expected
JURNI-BB-050	Report post	Saved
JURNI-BB-051	Duplicate report	Warning

4.3.8 Admin Panel

ID	Input	Expected
JURNI-BB-060	Review report	View details
JURNI-BB-061	Resolve	Status updated

4.4 SmokeTesting

ID	Test	Expected
JURNI-SM-001	App loads	Homepage visible
JURNI-SM-002	Register page loads	OTP form shows
JURNI-SM-003	Login works	Dashboard loads

4.5 Performance Testing

ID	Test	Expected
JURNI-PF-001	Homepage load	< 20 seconds
JURNI-PF-002	Media upload	< 50 seconds
JURNI-PF-003	Infinite scroll feed	Smooth scrolling

4.6 Stress Testing

ID	Stress Condition	Expected
JURNI-ST-001	10 users login at once	Server stable
JURNI-ST-002	15 posts uploaded	No crash
JURNI-ST-003	20 comments/min	No data corruption

5. Future Enhancements

1. AI Based Content Moderation

Implement automated detection of spam, harmful content, and inappropriate media.

2. Real-Time Chat System

Add instant messaging between users using WebSockets for live interactions.

3. Advanced Recommendation Engine

Suggest posts, users, and hashtags based on user activity and preferences.

4. Two Factor Authentication (2FA)

Strengthen login security with OTP or authenticator app verification.

5. Push Notification System

Notify users about likes, comments, follows, and reports in real time.

6. Multi Language Support

Offer the platform in multiple languages for global accessibility.

7. Creator Analytics Dashboard

Provide insights on profile reach, post performance, and audience engagement.

8. Scheduled Post Publishing

Allow users to schedule posts for future automatic publishing.

9. Mobile App Development

Launch native Android and iOS apps for better accessibility and user experience.

10. Report Auto Resolution System

Automatically categorize and resolve low-risk reports to reduce manual admin review.

6. Conclusion

This documentation has presented a comprehensive analysis, design, and implementation overview of the Travel Discovery & Social Sharing Platform (Jurni). The project was initiated to address the limitations of existing travel-related systems, which are fragmented, time-consuming, and lack authentic community-driven engagement. Through systematic requirement analysis, feasibility studies, and structured design approaches, the proposed system successfully demonstrates a unified solution that integrates travel discovery, social networking, and intelligent recommendations into a single platform.

The system analysis phase clearly identified the drawbacks of current travel platforms, such as the lack of integration, limited personalization, and absence of collaborative features. Based on these findings, the proposed system was designed with a strong focus on user experience, authenticity, and scalability. Key functionalities such as user authentication, profile management, content creation, social interactions (likes, comments, follows), reporting mechanisms, and admin moderation were carefully modeled using ERDs, UML diagrams, and system flow charts, ensuring clarity and correctness in system behavior and data movement.

The database design, supported by a detailed data dictionary, ensures structured data storage, integrity, and efficient retrieval using MongoDB. The separation of concerns between users, content, interactions, and administrative controls enhances system maintainability and security. Furthermore, the adoption of the Agile development model provides flexibility, continuous feedback, and incremental improvements, making the system adaptable to evolving user needs and future enhancements.

In conclusion, the Jurni Travel Discovery & Social Sharing Platform successfully fulfills its objectives by offering a scalable, user-centric, and feature-rich solution for modern travelers. It establishes a strong foundation for future advancements such as AI-powered itinerary planning, real-time communication, and mobile application development. Overall, this project demonstrates the effective application of software engineering principles to solve real-world problems and deliver a reliable, integrated digital travel experience.

7. Abbreviations

Sr. No.	Abbreviation	Full Form
1.	SRS	Software Requirements Specification
2.	AI	Artificial Intelligence
3.	API	Application Programming Interface
4.	JWT	JSON Web Token
5.	bcrypt	Password Hashing Library
6.	SPA	Single Page Application
7.	DBMS	Database Management System
8.	ORM	Object Relational Mapping
9.	DFD	Data Flow Diagram
10.	ERD	Entity Relationship Diagram
11.	IDE	Integrated Development Environment
12.	RAM	Random Access Memory
13.	CPU	Central Processing Unit
14.	CRUD	Create, Read, Update, Delete
15.	UAT	User Acceptance Testing
16.	SM	Smoke Testing
17.	PF	Performance Testing
18.	ST	Stress Testing
19.	BB	Black Box Testing
20.	WB	White Box Testing

8. References

8.1 Books & Academic References

1. Sommerville, Ian. *Software Engineering*, 10th Edition, Pearson Education.
2. Pressman, Roger S. *Software Engineering: A Practitioner's Approach*, McGraw-Hill.
3. Fowler, Martin. *UML Distilled: A Brief Guide to the Standard Object Modeling Language*.
4. Elmasri & Navathe. *Fundamentals of Database Systems*, Pearson.

8.2 Web Technologies Documentation

- Next.js Official Documentation
<https://nextjs.org/docs>
- React.js Official Guide
<https://react.dev>
- Node.js Documentation
<https://nodejs.org/en/docs>
- MongoDB Documentation
<https://www.mongodb.com/docs>
- Mongoose ODM Documentation
<https://mongoosejs.com/docs>
- Cloudflare R2 Storage Documentation
<https://developers.cloudflare.com/r2>
- JSON Web Token (JWT) Documentation
<https://jwt.io>
- bcrypt Password Hashing Documentation
<https://www.npmjs.com/package/bcrypt>

8.3 Design & Diagram Tools

- Draw.io
<https://app.diagrams.net>

8.4 Testing References

- Software Testing Fundamentals
<https://softwaretestingfundamentals.com>
- ISTQB Foundation concepts
<https://www.istqb.org>

8.5 Additional Learning & Support Resources

- MDN Web Docs (HTML, CSS, JS reference)
<https://developer.mozilla.org>
- Stack Overflow – Developer Discussions
<https://stackoverflow.com>
- YouTube – Traversy Media (Full Stack Tutorials)
<https://www.youtube.com/c/TraversyMedia>